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RESEARCH ARTICLE

MoLE: Mixture-of-LoRA Experts for Parameter-Efficient Dermoscopic Visual Question Answering

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ABSTRACT Dermoscopy is a non-invasive skin imaging technique that improves melanoma diagnosis by 10–30% compared with naked-eye examination. However, its use is limited by the extensive training required for dermoscopic interpretation. Most existing deep learning methods for dermoscopic image analysis are based on closed-set classification and do not support interactive consultation. Vision-language models can bridge this gap, but their application to dermatology is constrained by the limited availability of domain-specific data. We therefore propose a Mixture-of-LoRA-Experts (MoLE) architecture for dermoscopic visual question answering. MoLE is based on the pre-trained LLaVA-1.5-7B architecture and uses three independent Low-Rank Adaptation (LoRA) experts with a learnable router. The model was trained on approximately 1,200 ISIC dermoscopic images and 24,000 question–answer pairs produced by trained research annotators under a shared clinical rubric, with a dermatologist-reviewed subset. Without explicit task labels, the experts developed specializations in diagnostic, feature-detection, and general observational tasks. MoLE outperformed a single-LoRA model by 16.9% in METEOR and 19.1% in CIDEr. In a blinded study, three board-certified dermatologists rated MoLE responses significantly higher in clinical accuracy and domain-specific terminology. A cross-dataset evaluation on held-out HAM10000 images provided a near-domain robustness check. MoLE updated only 0.5% of the model's total parameters per forward pass and required 8 hours of training on a single 48 GB GPU. On an NVIDIA RTX 4070 Ti with 12 GB of memory, it achieved an average end-to-end latency of approximately 2.1 s per question and a generation throughput of approximately 28 tokens/s. These results demonstrate the viability of mixture-of-experts fine-tuning for dermatology. MoLE remains a research prototype intended for educational and second-opinion support under qualified supervision; it is neither validated nor intended for direct patient-facing use.

INDEX TERMS
Mixture-of-experts, vision–language models, dermoscopy, low-rank adaptation, parameter-efficient fine-tuning, medical visual question answering, expert specialization.

I. INTRODUCTION

Melanoma represents one of the most dangerous types of skin cancer. It arises from melanocytes in response to prolonged UV light exposure. The 5-year survival rate for patients diagnosed with melanoma when it is still in its local or regional (early) stage is greater than 99%. However, when it spreads (metastasis), the survival rate of these individuals falls to about 27% [1], [2]. The high mortality rates of melanoma are due to the fact that this disease is

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difficult to diagnose early and accurately. A study found that dermoscopy (which uses light to view and magnify the skin layers below the surface) can increase melanoma detection by 30% [3]. Dermoscopy does provide an advantage in terms of improved identification of melanomas, however, it requires advanced training that may not be available in limited resource-based facilities.

Advances in artificial intelligence have enabled deep learning models to narrow some gaps in specialist image interpretation. Convolutional neural networks (CNNs) have achieved strong performance on dermatologic classification benchmarks [4], [5]. However, closed-set classifiers generally provide neither detailed qualitative feedback nor interactive dialogue. Vision-language models (VLMs) [6], [7] enable natural-language interaction with images, but their scale creates two challenges in a specialized domain such as dermoscopy. First, annotated dermoscopic data for open-ended question answering are limited. Second, dermoscopic decision-making involves several reasoning modes. A single consultation may require recognition of morphologic features, systematic visual description, and integration of those findings into a differential diagnosis. A useful model must therefore support these reasoning modes simultaneously.

Low-Rank Adaptation (LoRA) [8] provides an efficient way to adapt a pre-trained model under limited-data conditions by adding trainable low-rank matrices to selected layers. However, typical single-adapter methods apply the same update path to every input, which is a poor fit for the diverse reasoning required in dermoscopic visual question answering. Mixture-of-Experts (MoE) architectures [9], [10], [11] address heterogeneous inputs by routing each input to a specialized submodel. Combining MoE routing with LoRA adapters can therefore provide both parameter efficiency and task-specific adaptation while leaving the base-model weights frozen.

We developed a Mixture-of-LoRA-Experts (MoLE) architecture for dermoscopic visual question answering (VQA). A learned router dynamically assigns each image–question pair to one of three independent LoRA experts, enabling distinct adaptations for diagnostic, feature-oriented, and descriptive reasoning without explicit task labels.

While the individual ingredients of MoLE—LoRA, sparse routing, and vision-language fine-tuning—are established, their application to label-free specialization within a single clinical VQA domain remains underexplored. Earlier LoRA-composition and MoE-style plugin methods [12], [13] organize experts around separately trained adapters, predefined task groups, or distinct data distributions. In contrast, the experts in our system specialize within dermoscopy according to the reasoning mode required by each question, and the partition emerges under a load-balancing objective without task labels. We evaluate this intra-domain specialization through routing analysis, parameter-matched controls, ablations, and blinded clinician assessment.

The MoLE architecture integrates expert routing with parameter-efficient fine-tuning for vision-language models while remaining compatible with consumer-grade hardware. We constructed a dermoscopic VQA dataset of 1,200 ISIC images and 24,000 question–answer pairs annotated by trained research engineers; two board-certified dermatologists reviewed a stratified 15% subset. Expert specialization emerged during load-balanced training without explicit routing labels. We evaluate MoLE through ablations on the number of experts, LoRA rank, and routing strategy. Additional analyses include blinded assessment by board-certified dermatologists, error analysis stratified by lesion category, safety and calibration analysis, and a near-domain cross-dataset evaluation on HAM10000.

On a test set of 180 unseen images (3,600 QA pairs), MoLE achieves a 16.9% improvement in METEOR and a 19.1% improvement in CIDEr over a matched single-LoRA baseline. A blinded evaluation by three board-certified dermatologists confirms statistically significant gains in clinical accuracy ($p < 0.01$, paired t-test). These results indicate that MoLE can capture heterogeneous reasoning patterns required for dermoscopic interpretation while remaining deployable on modest hardware. As a research prototype, MoLE is intended to support trained clinicians and learners by providing structured descriptions and differential considerations in an educational or second-opinion capacity. It is not intended for direct patient-facing use, and any output must be interpreted by a qualified dermatologist within the full clinical context. The safety mechanisms discussed in this article, including specialist-referral prompts, deterministic high-risk keyword checks, and low-routing-confidence escalation, are proposed safeguards that require prospective validation and are not presented as clinically established.

II. RELATED WORK

A. VISION–LANGUAGE MODELS

Vision-language models have evolved through several architectural paradigms. Early models, including VisualBERT [14] and LXMERT [15], used cross-modal transformers and co-attention mechanisms to produce joint text–image representations, but at substantial computational cost. CLIP [16] subsequently demonstrated that contrastive training on 400 million web-collected image–text pairs could produce transferable visual representations that support new tasks without task-specific fine-tuning. BLIP [17] introduced a bootstrapping cycle in which synthetic captions are generated, quality-filtered, and used to augment the original training data. BLIP-2 [18] further reduced training cost by placing a lightweight query transformer, Q-Former, between frozen vision encoders and frozen language models.

LLaVA simplified this design by connecting a CLIP visual encoder to a Vicuna language model through a learned projection layer. Instruction tuning on GPT-4-generated descriptions achieved competitive performance with lower architectural complexity. LLaVA-1.5 [19]

improved high-resolution image encoding and visual tokenization. This comparatively simple architecture provides a practical basis for domain-specific adaptation.

LLaVA-Med [7] adapted LLaVA for biomedical tasks through fine-tuning on medical image-text pairs and showed improvements on radiology and pathology benchmark tasks. Med-Flamingo [20] also preserved few-shot learning abilities for medical context. Domain-specific models such as RadFM [21] (radiology), CheXagent [22] (chest X-ray), and PathChat [23] (pathology) have shown that targeted data and architectural modifications improve performance on specialized medical tasks. BiomedCLIP [24] has also provided domain-adapted visual representations for biomedical applications. Dermoscopy remains relatively under-explored compared with other medical domains. Most existing work is limited to classification, and prior systems do not support the heterogeneous reasoning required for interactive dermoscopic VQA. MoLE addresses this gap through parameter-efficient adaptation of a general vision-language architecture.

B. PARAMETER-EFFICIENT FINE-TUNING

Training billion-parameter models for medical applications requires substantial computing resources and large datasets, both of which are often limited in specialized clinical domains. Parameter-efficient fine-tuning (PEFT) addresses this challenge by updating only a small fraction of model parameters. LoRA [8] introduces trainable low-rank matrices and can reduce the number of trainable parameters by several orders of magnitude while approaching full fine-tuning performance. It constrains adaptation to a low-dimensional update space and leaves the original model weights unchanged.

QLoRA [25] extends LoRA by quantizing the base model to 4-bit NormalFloat4 (NF4), a format designed for normally distributed weights. Combined with paged optimizers, QLoRA enables fine-tuning of models with up to 65 billion parameters on commodity GPUs.

Several alternatives to PEFT have been developed. Prefix-tuning [26] adds learnable vectors (which are real numbers) to each layer of a transformer network before learning. Prompt-tuning [27] simply learns the input embeddings. Adapter-modules [28] add lightweight "bottleneck" networks between transformer layers in which knowledge is stored. Among these, LoRA remains the most widely adopted for large language models due to its simplicity, strong performance, and compatibility with quantization.

However, single-adapter methods apply the same update path regardless of input characteristics. In dermoscopic VQA, one adapter may be insufficient to capture the full range of diagnostic, feature-oriented, and descriptive capabilities. This limitation motivates a mixture-of-experts architecture with separate adapters for different reasoning modes.

C. MIXTURE-OF-EXPERTS

Mixture-of-Experts (MoE) methods divide the computations into multiple specialized sub-models using learned routing [9], [10]. The routing was made simpler with the Switch Transformer [11] by directing each token to a single expert so that the communication overhead could be reduced. Training MoE at scale was made possible by extending MoE with the GShard method [29].

The two major challenges in MoE training are expert collapse, in which routing concentrates on a few experts, and load imbalance from skewed routing distributions. Load-balancing losses [11] penalize uneven expert utilization, while router z-loss [30] stabilizes training by penalizing large routing logits. ST-MoE [30] showed that the proper initialization and regularization of the routing mechanism are critical for the successful learning of the experts. Recent innovations include expert-choice routing [31] and dynamic assignment based on task difficulty.

The combination of MoE routing and PEFT is a comparatively recent research direction. Mixture-of-LoRA methods [12] learn to combine specialized adapters, while LoRAMoE [13] applies an MoE-style LoRA plugin to mitigate world-knowledge forgetting during instruction tuning. We extend this direction to vision-language models for dermoscopic analysis, where expert specialization emerges from heterogeneous reasoning requirements within one domain rather than from predefined task groups.

D. AI IN DERMOSCOPY

Deep learning has influenced computer-aided diagnosis across several medical-imaging modalities. In radiology, pathology, and chest imaging, vision-language and foundation models such as LLaVA-Med, RadFM, CheXagent, and PathChat have been developed to support image interpretation and clinical-language generation [7], [21], [22], [23]. These systems illustrate a broader progression from task-specific detection or segmentation toward interactive, multimodal clinical assistance. In dermoscopy, Esteva et al. [5] trained an Inception-v3 model on more than 129,000 clinical images and reported performance comparable to that of 21 board-certified dermatologists. Haenssle et al. [4] compared a deep CNN with 58 dermatologists from 17 countries and reported high sensitivity and specificity. Attention mechanisms [32] can improve classification while highlighting visually relevant regions, and multi-task approaches [33] can jointly model diagnoses and dermoscopic characteristics. The ISIC Archive [34] and HAM10000 [35] are widely used benchmarks for reproducible dermoscopic-image analysis.

Despite substantial progress in automatic skin-lesion analysis, most dermoscopic systems remain limited to lesion classification. They neither explain their decisions in free-form clinical language nor respond to follow-up questions. Vision-language models can address these limitations by enabling interactive analysis of medical images. Our results show

that MoE architectures can capture several reasoning modes involved in dermoscopic VQA while remaining efficient enough for widely available hardware.

III. METHODOLOGY

This section describes the dermoscopic VQA system and is organized as follows: (A) Problem Formulation, (B) Dataset Construction, (C) Base Model Architecture, (D) Mixture-of-LoRA-Experts Architecture, and (E) Implementation Details.

A. PROBLEM FORMULATION

Dermoscopic VQA can be formulated formally as a conditional generation problem. Given an input image $\mathbf{I} \in \mathbb{R}^{H\times W\times 3}$ and a question token sequence $\mathbf{x} = [x_1, x_2, \ldots, x_T]$, the model generates an answer sequence $\mathbf{y} = [y_1, y_2, \ldots, y_S]$. In other words, it learns the conditional probability distribution:

$$P(\mathbf{y}|\mathbf{I}, \mathbf{x}; \theta) \tag{1}$$

where θ denotes the model's learnable parameters. The autoregressive generation process is

$$P(\mathbf{y}|\mathbf{I}, \mathbf{x}) = \prod_{s=1}^{S} P(y_s|\mathbf{I}, \mathbf{x}, y_{<s}; \theta) \tag{2}$$

where each token is conditioned on the image, question, and all previously generated tokens.

The biggest challenge in dermoscopic VQA is the qualitative differences in reasoning involved in the various question types. To account for the qualitative differences, we use multi-adapter architecture with dynamic routing that selects the best performing adapter based on the type of input.

B. DATASET CONSTRUCTION

1) IMAGE SELECTION AND DIAGNOSTIC DISTRIBUTION

We collected 1,200 dermoscopic images from the International Skin Imaging Collaboration (ISIC) Archive [34]. Images were selected to represent diverse visual features, diagnoses, and available patient demographics. The dataset contains 840 benign lesions (70%), 288 malignant lesions (24%), and 72 indeterminate lesions (6%). Malignant lesions were intentionally oversampled relative to the estimated 5–10% biopsy malignancy rate to provide sufficient examples of pathologic features. The benign group comprises 432 melanocytic nevi, 168 seborrheic keratoses, 120 dermatofibromas, and 120 hemangiomas. The malignant group comprises 120 basal cell carcinomas, 108 melanomas, and 60 squamous cell carcinomas. Most indeterminate cases are solar or actinic keratoses.

2) ANNOTATION PROCESS AND QUALITY ASSURANCE

Four trained research engineers performed the primary annotation using a shared rubric and controlled answer formats. A senior annotator supervised consistency and resolved disagreements. Before full-scale labeling, the team completed two pilot calibration rounds, each comprising 50 images and 1,000 QA pairs, to standardize interpretation of ABCDE-informed descriptors, common dermoscopic terms, and acceptable answer formats. For formal quality control, a stratified 20% subset was independently double-annotated. The senior annotator resolved all disagreements according to predefined uncertainty rules, including the use of *indeterminate* or *insufficient visual evidence*. Inter-annotator agreement was substantial (Cohen's $\kappa = 0.81$).

In addition, two board-certified dermatologists reviewed a stratified random 15% subset (180 images; 3,600 QA pairs) to assess rubric adherence, identify clinically ambiguous or unsafe phrasing, and detect systematic error patterns. Their review led to corrections in 461 QA pairs, corresponding to 12.8% of the reviewed subset and 1.92% of the complete 24,000-pair dataset. Corrections addressed clinical terminology, ambiguity, safety language, and answer structure. The resulting guidance was incorporated into the annotation rubric before the remaining annotations were completed; those remaining pairs did not receive item-level dermatologist review. Figure 1 illustrates the diversity of the three diagnostic categories.

3) QUESTION–ANSWER PAIR GENERATION

We created 20 question–answer pairs per image resulting in a total of 24,000 pairs. These questions were designed to cover seven reasoning types, as summarized in Table 1.

TABLE 1. Distribution of question types in the dataset.

Question Type	Count	Percentage
Diagnosis	2,280	9.5%
Feature Detection	4,416	18.4%
Visual Observation	7,704	32.1%
Pattern Description	1,968	8.2%
Body Site	912	3.8%
General Description	1,536	6.4%
Yes/No Questions	5,184	21.6%

The majority of questions are visual observation questions because systematic evaluation of morphology is a basic component of dermoscopic analysis. Feature-detection questions are also common and they cover morphological structures such as pigment networks, globules, streaks, and vascular patterns. Diagnostic questions have lower numbers of cases. However, they represent the higher degree of analytical reasoning.

4) ANSWER QUALITY GUIDELINES

All answers were created using an established set of quality guidelines. Those guidelines include correct dermatoscopic terminology usage along with adequate description. The guidelines also require answers to clearly indicate when an image cannot lead to a specific diagnosis (i.e., "this lesion has characteristics of melanoma. However, clinical correlation and possible biopsy are required before the final diagnosis."). The answers to complex questions follow a structural format. The first section describes the observable features, the second

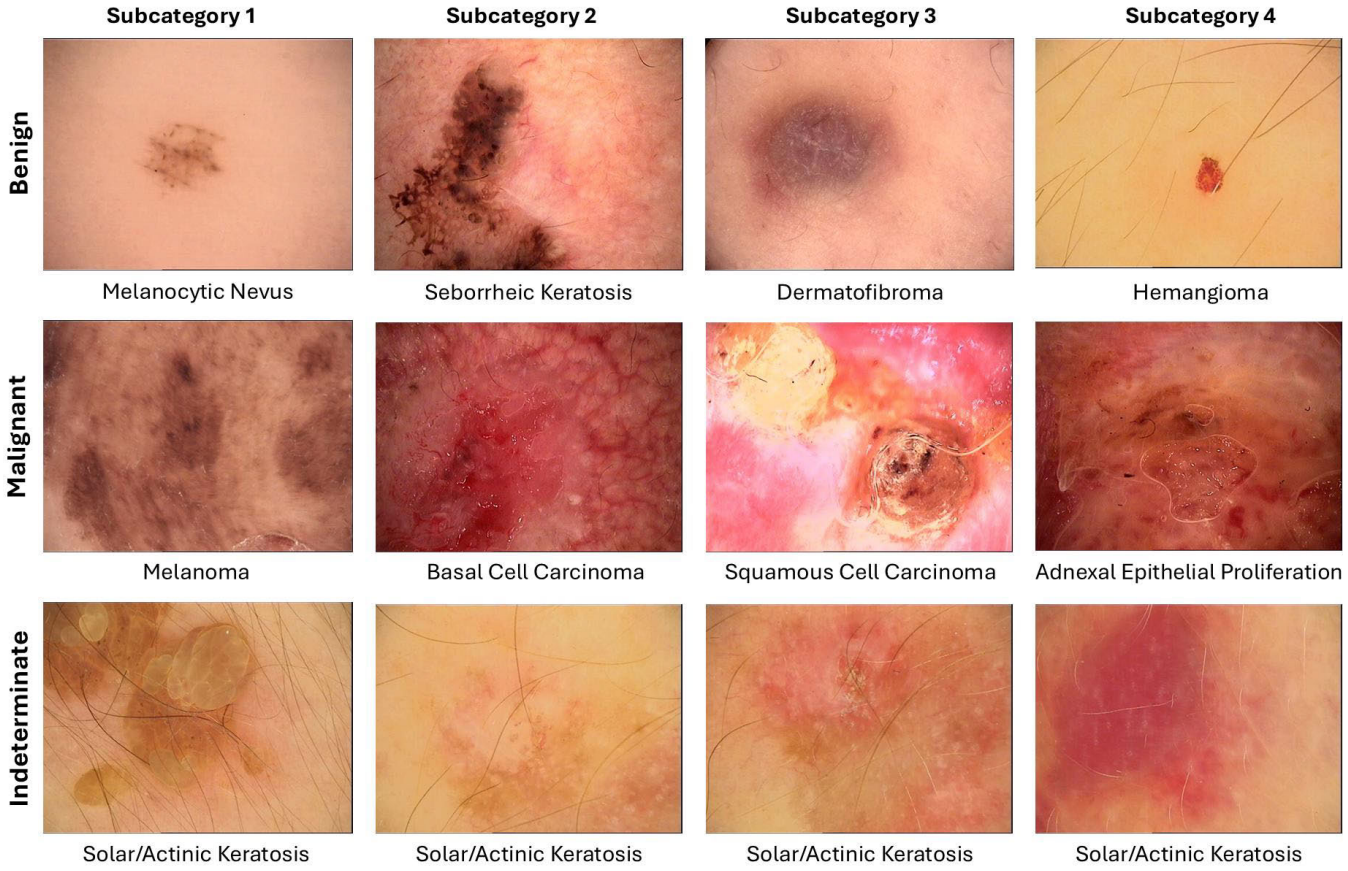


FIGURE 1. Representative dermoscopic images across diagnostic categories: benign (top), malignant (middle), and indeterminate (bottom).

section discusses the reassurance provided by those features, and the third section provides a diagnostic impression. Consistent terminology was used across images to prevent model confusion from identical findings.

5) DATA SPLITTING STRATEGY

We split the dataset at the image level into 85% training (1,020 images; 20,400 QA pairs) and 15% test (180 images; 3,600 QA pairs). All 20 questions for a given image appear entirely in either the training or test set, ensuring evaluation on truly unseen images rather than on new questions about previously observed images. Diagnostic and question-type distributions were balanced across both splits.

6) TRAIN–TEST OVERLAP ANALYSIS

Because the QA pairs are built from a shared set of clinical question families, we quantified the lexical and template overlap between the training and test splits to verify that reported performance is not inflated by leakage (Table 2). The image-level split is exact: no image and no verbatim question–answer pair is shared between the two partitions. As expected, question *templates* are largely shared—95.9% of test questions use a template also seen in training, and short question n-grams overlap heavily (1-gram 0.891, 2-gram 0.666)—because the dataset deliberately uses a controlled set of clinical question types. This is by design and does not constitute leakage, since the answer depends on image-specific visual findings. Critically, overlap in the *answers* falls off sharply with n-gram order (1-gram 0.446, 2-gram 0.189, 3-gram 0.080, 4-gram 0.030), and only 2.5% of test answers share more than 20% of their 4-grams with their nearest training answer. Test answers are therefore not lexically copied from training; they describe genuinely unseen images, supporting the validity of the reported metrics.

TABLE 2. Train–test overlap checks (20,400 train; 3,600 test QA pairs).

Check	Rate
Images shared across train and test	0.000
Exact duplicate QA pairs across splits	0.000
Test questions using a train-seen template	0.959
Test answers with >0.20 nearest 4-gram overlap	0.025
Question text 1-gram / 2-gram overlap	0.891 / 0.666
Question text 3-gram / 4-gram overlap	0.444 / 0.278
Answer text 1-gram / 2-gram overlap	0.446 / 0.189
Answer text 3-gram / 4-gram overlap	0.080 / 0.030

C. BASE MODEL ARCHITECTURE

We build on LLaVA-1.5-7B [19], a vision-language model composed of a pretrained vision encoder and a language model connected through a learned projection layer.

1) VISION–LANGUAGE INTEGRATION

LLaVA-1.5 couples a CLIP Vision Transformer (ViT-L/14) [16] with the Vicuna-7B language model [36] via a two-layer MLP projection. The visual encoder processes input images resized to 224×224 pixels, dividing them into 14×14 non-overlapping patches of size 16×16. Each patch is linearly projected into a 1024-dimensional embedding, and the CLIP ViT-L/14 processes these embeddings through 24 transformer layers with 16 attention heads, producing a sequence of visual feature vectors $\mathbf{V} = [\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_N] \in \mathbb{R}^{N \times 1024}$, where $N = 196$.

The two-layer MLP projection bridges the dimensionality mismatch between the vision encoder (1024-dimensional) and the language model (4096-dimensional):

$$\mathbf{H}_v = \mathbf{W}_2 \cdot \text{GELU}(\mathbf{W}_1 \cdot \mathbf{V}^T + \mathbf{b}_1) + \mathbf{b}_2 \tag{3}$$

where $\mathbf{W}_1 \in \mathbb{R}^{d_{hidden} \times 1024}$ and $\mathbf{W}_2 \in \mathbb{R}^{4096 \times d_{hidden}}$. This projection is learned during LLaVA's pretraining on general instruction-following tasks.

2) AUTOREGRESSIVE GENERATION

The Vicuna-7B language decoder is a 32-layer transformer with causal self-attention, trained on user-shared conversations for conversational competency. Projected visual features are prepended to text token embeddings, forming a combined input $[\mathbf{H}_v, \mathbf{E}_x]$ where $\mathbf{E}_x \in \mathbb{R}^{T \times 4096}$. The causal attention mask ensures text tokens can attend to all visual tokens and preceding text tokens but not future text tokens, yielding the autoregressive factorization:

$$P(\mathbf{y}|\mathbf{I}, \mathbf{x}) = \prod_{s=1}^{S} P(y_s|\mathbf{H}_v, \mathbf{x}, y_{<s}; \theta) \tag{4}$$

3) LIMITATIONS OF SINGLE-ADAPTER FINE-TUNING

While LLaVA provides capable visual reasoning, adapting it to dermoscopy with a single LoRA adapter [8] applies uniform weight updates across all inputs. This uniformity is suboptimal for dermoscopic VQA, where diagnostic, feature-detection, and descriptive questions engage distinct reasoning pathways. Our MoLE architecture overcomes this limitation by learning multiple specialized adaptations with dynamic routing.

D. MIXTURE-OF-LoRA-EXPERTS ARCHITECTURE

The MoLE architecture, illustrated in Figure 2, applies multiple independent LoRA experts to the language model with a learned router that assigns each input to a single expert. The visual encoder and projection layer remain shared and frozen across all experts.

1) DESIGN RATIONALE

The MoLE architecture has been designed based on three main considerations. Only the language model is subjected to expert specialization, as both the vision encoder and the projection layer are shared and frozen. This reduces the number of parameters and avoids the potential for overfitting to a small dataset. The number of experts is fixed at $K = 3$, and this decision is validated in the ablation study (Section IV-H1). The router uses sparse top-1 selection which means every input is processed by a single expert. This maximizes the computational efficiency at inference time.

2) LOW-RANK EXPERT ADAPTATIONS

Following LoRA [8], each expert parameterizes its weight updates as low-rank matrix decompositions. For a pretrained weight matrix $\mathbf{W}_0 \in \mathbb{R}^{d \times k}$, expert e learns:

$$\Delta\mathbf{W}^{(e)} = \mathbf{B}^{(e)}\mathbf{A}^{(e)} \tag{5}$$

where $\mathbf{B}^{(e)} \in \mathbb{R}^{d \times r}$ and $\mathbf{A}^{(e)} \in \mathbb{R}^{r \times k}$ with rank $r \ll \min(d, k)$. The forward pass becomes:

$$\mathbf{h} = \mathbf{W}_0\mathbf{x} + \frac{\alpha}{r}\mathbf{B}^{(e)}\mathbf{A}^{(e)}\mathbf{x} \tag{6}$$

where α is a scaling hyperparameter.

LoRA is applied to seven weight matrices per transformer layer: query ($\mathbf{W}_q$), key ($\mathbf{W}_k$), value ($\mathbf{W}_v$), and output ($\mathbf{W}_o$) projections in the attention module, plus gate ($\mathbf{W}_{gate}$), up ($\mathbf{W}_{up}$), and down ($\mathbf{W}_{down}$) projections in the feed-forward network. With 32 layers, rank $r = 16$, and three experts, this yields approximately 29 million trainable parameters per expert (87 million total for expert adaptations, excluding the router).

Each expert maintains independent low-rank factors $\{\mathbf{B}^{(e)}, \mathbf{A}^{(e)}\}$ for each adapted layer. The matrix $\mathbf{A}^{(e)}$ is initialized from a Gaussian distribution scaled by $1/\sqrt{r}$, while $\mathbf{B}^{(e)}$ is initialized to zero, ensuring all experts begin at the pretrained state.

3) ROUTER NETWORK

The router determines expert assignment for each input. Given token embeddings $\mathbf{E}_x \in \mathbb{R}^{T \times d}$ for a question of length T with $d = 4096$, the router first computes a fixed-length representation via mean pooling:

$$\mathbf{r} = \frac{1}{T}\sum_{t=1}^{T}\mathbf{E}_x[t] \in \mathbb{R}^d \tag{7}$$

This representation is processed through a three-layer feedforward network:

$$\begin{aligned} \mathbf{h}_1 &= \text{ReLU}(\text{LayerNorm}(\mathbf{W}_1\mathbf{r} + \mathbf{b}_1)) \\ \mathbf{h}_2 &= \text{ReLU}(\mathbf{W}_2\mathbf{h}_1 + \mathbf{b}_2) \\ \mathbf{z} &= \mathbf{W}_3\mathbf{h}_2 + \mathbf{b}_3 \end{aligned} \tag{8}$$

where $\mathbf{W}_1 \in \mathbb{R}^{512 \times 4096}$, $\mathbf{W}_2 \in \mathbb{R}^{256 \times 512}$, $\mathbf{W}_3 \in \mathbb{R}^{K \times 256}$, and K is the number of experts. Dropout with probability 0.1 is applied after each ReLU activation.

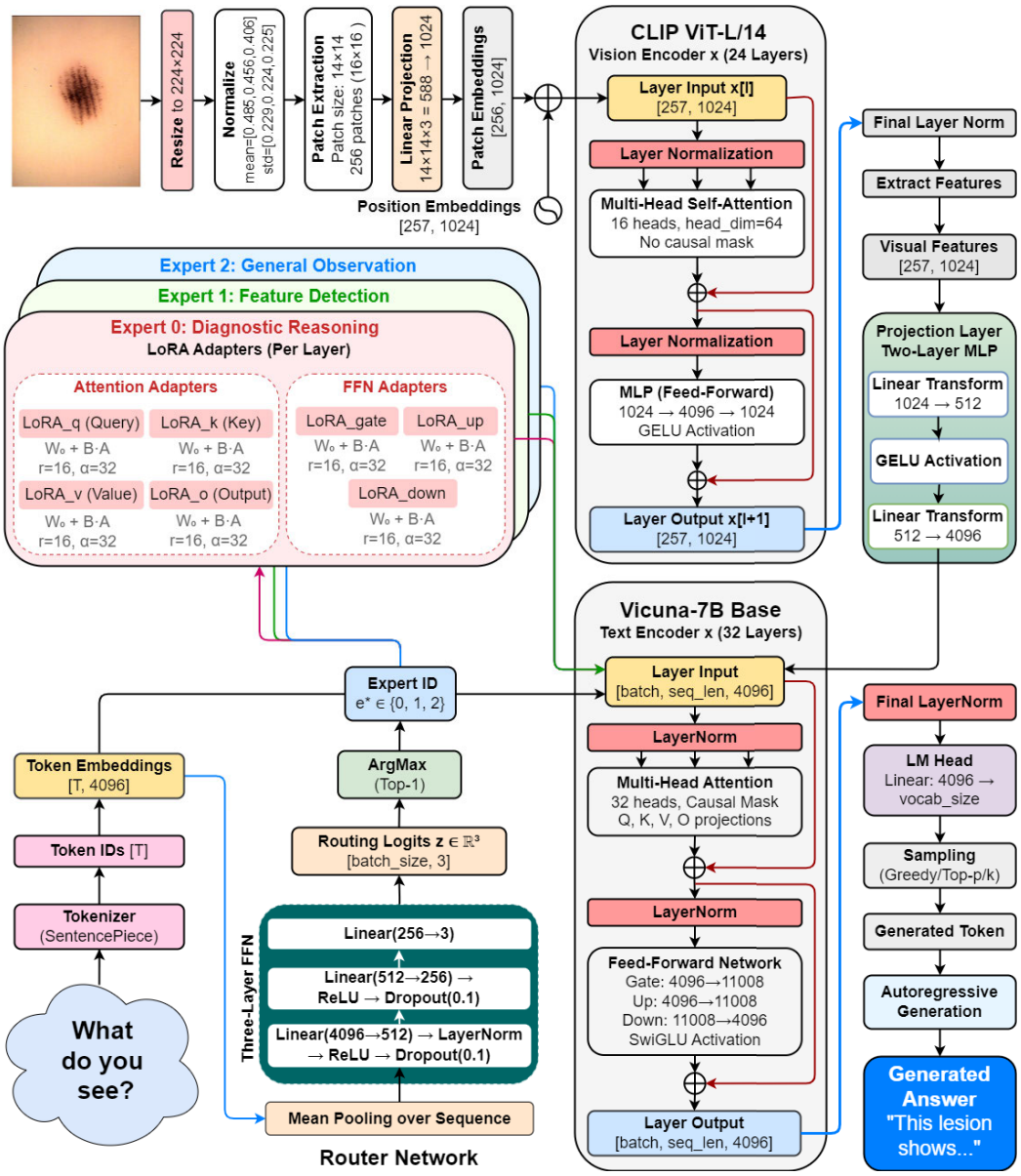


FIGURE 2. Overview of the MoLE architecture with three LoRA experts, learned router, and shared vision encoder.

Softmax converts the logits $\mathbf{z} \in \mathbb{R}^K$ to routing probabilities:

$$p_e = \frac{\exp(z_e)}{\sum_{e'=1}^{K} \exp(z_{e'})} \tag{9}$$

Top-1 routing assigns each input to the expert with the highest probability:

$$e^* = \underset{e \in \{1,\ldots,K\}}{\operatorname{argmax}}\ p_e \tag{10}$$

Router gradients are computed through the softmax probabilities while expert selection uses the straight-through estimator, preventing routing decisions from interfering with expert parameter learning.

4) TRAINING OBJECTIVE

The training objective combines language modeling loss with auxiliary losses for routing stability. For a batch $\{\mathbf{I}_i, \mathbf{x}_i, \mathbf{y}_i\}_{i=1}^{B}$, the language modeling loss is:

$$\mathcal{L}_{LM} = -\frac{1}{B} \sum_{i=1}^{B} \sum_{s=1}^{S_i} \log P(y_{i,s} | \mathbf{H}_v^{(i)}, \mathbf{x}_i, \mathbf{y}_{i,<s}; \theta^{(e_i^*)}) \tag{11}$$

where e_i^* is the expert assigned to example i.

A load-balancing loss prevents routing collapse:

$$\mathcal{L}_{balance} = \sum_{e=1}^{K} \left(\bar{p}_e - \frac{1}{K} \right)^2 \tag{12}$$

where $\bar{p}_e = \frac{1}{B}\sum_{i=1}^{B} p_e^{(i)}$ is the mean routing probability for expert e across the batch.

A router z-loss [30] stabilizes training by penalizing large logit magnitudes:

$$\mathcal{L}_z = \frac{1}{B}\sum_{i=1}^{B}\left(\log\sum_{e=1}^{K}\exp(z_e^{(i)})\right)^2 \tag{13}$$

The combined objective is:

$$\mathcal{L} = \mathcal{L}_{LM} + \lambda_{balance}\mathcal{L}_{balance} + \lambda_z\mathcal{L}_z \tag{14}$$

with $\lambda_{balance} = 0.001$ and $\lambda_z = 0.0001$.

E. IMPLEMENTATION DETAILS

1) SOFTWARE FRAMEWORK AND MODEL QUANTIZATION

MoLE is implemented in PyTorch 2.5.1 with Hugging Face Transformers (v4.36.0) and PEFT for LoRA integration. The base LLaVA-1.5-7B model is loaded with 4-bit NF4 quantization via QLoRA [25], reducing its memory footprint to approximately 4.5 GB. Base model weights remain frozen in 4-bit precision while gradients and optimizer states use bfloat16 for numerical stability.

2) ARCHITECTURE CONFIGURATION

We configure three LoRA experts ($K = 3$) with rank $r = 16$, scaling factor $\alpha = 32$, and LoRA dropout $p = 0.05$. The rank was selected based on the ablation study in Section IV-H2. Independent expert adapters are created using PEFT's multi-adapter functionality, ensuring separate trainable parameters for each expert.

The router network uses hidden dimensions of 512 and 256 with dropout probability 0.1 after each ReLU activation, compressing 4096-dimensional question embeddings to K-dimensional expert logits.

3) TRAINING PROCEDURE

Training proceeds for 3 epochs with a batch size of 32, yielding approximately 1,913 optimization steps over the 20,400 training examples. We use AdamW optimization with a learning rate of 5×10^{-5} for expert parameters and 2.5×10^{-4} for router parameters. The $5\times$ higher router learning rate accelerates the differentiation of experts during early training when all adapters are near-identical. The learning rate follows a cosine annealing schedule with 300 warmup steps.

Gradient clipping with a maximum norm of 1.0 prevents training instability. Weight decay of 0.01 is applied to expert parameters; no weight decay is applied to the router, as the load-balancing loss already regularizes its behavior. The auxiliary loss weights $\lambda_{balance} = 0.001$ and $\lambda_z = 0.0001$ are intentionally small to allow specialization to emerge from the data distribution rather than being forced by the loss functions.

4) INFERENCE CONFIGURATION

At inference time, the router selects the single highest-probability expert for each input (top-1 routing) without blending or soft routing. Greedy decoding (selecting the highest-probability token at each step) is used for all models to ensure fair comparison.

5) COMPUTATIONAL REQUIREMENTS

Training completes in approximately 8 hours on a single NVIDIA L40S GPU (48 GB VRAM). Peak memory usage is 38 GB, comprising the quantized base model (4.5 GB), expert parameters in bfloat16 (∼350 MB total), optimizer states, and activation caches. At inference time, the quantized base model and all three expert adapters fit within 12 GB VRAM, enabling deployment on consumer-grade GPUs such as the NVIDIA RTX 4070 Ti.

6) BASELINE AND COMPARISON METHODS

We compare MoLE against six baselines trained and evaluated on the same data splits. (1) **Base LLaVA-1.5-7B** is the pretrained model evaluated zero-shot, establishing lower-bound performance. (2) **LLaVA-Med** [7] is the biomedical-adapted LLaVA variant, evaluated zero-shot on our dermoscopic test set to assess transfer from general medical pretraining. (3) **Single-LoRA** uses a single LoRA adapter with rank $r = 16$ and identical hyperparameters to a single MoLE expert, representing standard PEFT. (4) **Single-LoRA (3×r)** uses a single LoRA adapter with rank $r = 48$, matching the total parameter count of MoLE's three experts, to control for the possibility that MoLE's gains stem from increased parameter capacity alone. (5) **LLaVA-Med + LoRA** fine-tunes LLaVA-Med with a single dermoscopic LoRA adapter under exactly our training protocol (same rank, schedule, and data splits), providing a training-symmetric medical-pretraining comparison rather than a zero-shot one. (6) **All-expert ensemble** instantiates the same three trained LoRA experts but replaces the learned router with equal-weight averaging of all expert outputs (no routing), isolating the contribution of learned routing from the contribution of having multiple adapters.

All baselines use identical evaluation protocols, including greedy decoding and the same test split.

IV. RESULTS

This section presents experimental results organized as follows: training dynamics, expert specialization, routing behavior, performance comparisons with statistical significance testing, additional controlled baselines, per-question-type analysis, ablation studies (including vision-encoder freezing, router design, and expert collapse), error analysis stratified by lesion category, structured clinical metrics, safety and uncertainty analysis with calibration, clinician evaluation, cross-dataset validation, expert-specific performance, and qualitative examples.

A. TRAINING DYNAMICS

During training, the training loss decreased from 2.91 and then gradually converged to 0.74. The load-balancing loss also dropped from 0.0038 to 0.0004 representing a uniform

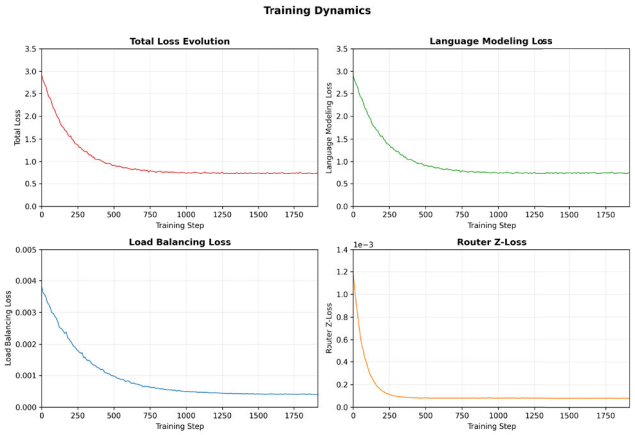


FIGURE 3. Evolution of training loss, language modeling loss, load-balancing loss, and router z-loss during MoLE training.

distribution of examples across experts. The router z-loss remained consistently below 0.0002 throughout the training. The various losses for each component are shown in Figure 3.

The mean training losses for Experts 0, 1, and 2 were 0.712, 0.738, and 0.761, respectively. These values reflect differences in the difficulty of the routed training examples.

B. EXPERT SPECIALIZATION

Table 3 gives the specialization of each expert in terms of question type over the test set. Expert 0 answers 41.8% of all examples and has a strong preference toward diagnostic questions answering 71.2% of them. Expert 1 answers 35.1% of examples and has a strong preference toward feature-detection questions (55.8%). Expert 2 answers the remaining 23.1% of examples and has a relatively strong preference toward visual-observation questions (33.6%).

TABLE 3. Expert usage distribution across question types on the test set (3,600 QA pairs).

Question Type	Expert 0	Expert 1	Expert 2	Total
Diagnosis	71.2%	19.4%	9.4%	342
Feature Detection	30.5%	55.8%	13.7%	662
Visual Observation	37.4%	29.0%	33.6%	1,156
Pattern Description	46.8%	35.1%	18.1%	295
Body Site	40.6%	33.3%	26.1%	137
General Description	38.3%	35.2%	26.5%	230
Yes/No	43.6%	34.8%	21.6%	778
Overall	**41.8%**	**35.1%**	**23.1%**	**3,600**

This specialization emerged automatically during load-balanced training without any routing labels. It shows that the router learns to partition the input space according to the reasoning requirements of each question type.

Figure 4 provides a summary of the specialization patterns.

C. ROUTING EVOLUTION

At the start of the training, the router randomly assigned samples to each of the three experts 0 (30%), 1 (33%), and 2 (37%). By the end, the expert distribution shifted to 42%, 35%, and 23% respectively. This early convergence shows that the router found an effective assignment strategy within the first epoch and then continued to refine it through additional epochs. The complete evolution of expert usage is shown in Figure 5.

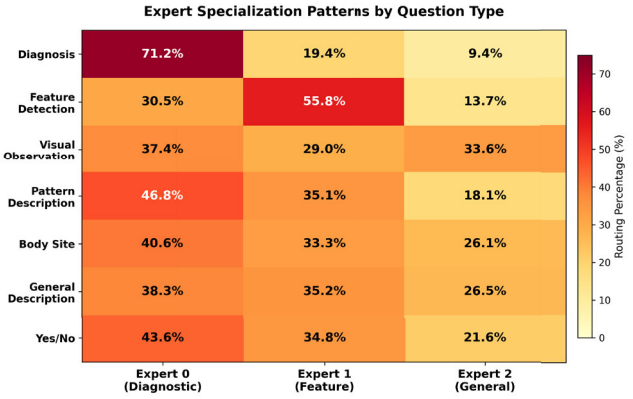


FIGURE 4. Expert specialization patterns across question types.

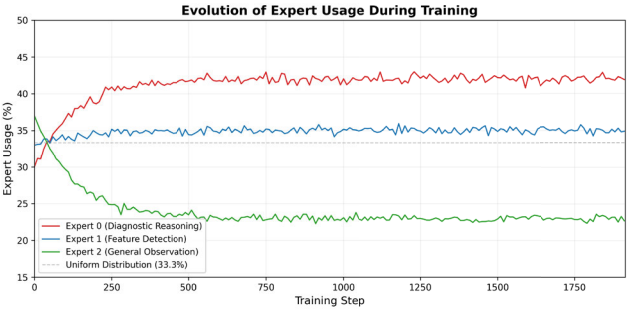


FIGURE 5. Evolution of expert usage percentages during training.

D. ROUTING CONFIDENCE

The mean maximum routing probability is 0.748, showing confident expert assignments. These routing confidence statistics are summarized in Table 4. The routing entropy averages 0.61 bits, compared with 1.585 bits for a uniform distribution over three experts, with a normalized confidence of 61.5%. The router assigns probability above 0.8 to the chosen expert in 41.3% of cases, while ambiguous routing (where no expert exceeds 0.5 probability) occurs in only 14.6% of cases.

TABLE 4. Router confidence statistics on the test set.

Metric	Value
Mean Max Probability	0.748
Routing Entropy (bits)	0.610
Max Entropy (bits)	1.585
Normalized Confidence	61.5%
Examples with $p > 0.8$	41.3%
Examples with $p > 0.5$	85.4%

E. PERFORMANCE COMPARISON

Table 5 compares MoLE against all baselines on the test set. All metrics are reported as mean ± 95% bootstrap confidence interval (1,000 resamples). Statistical significance is assessed using a paired bootstrap test [37] comparing MoLE against the Single-LoRA baseline. Because multiple metrics are compared simultaneously, we control the family-wise error rate using the Holm–Bonferroni procedure across the seven metrics in Table 5 (and, separately, across the three dimensions of the clinician evaluation in Section IV-L). All differences reported as significant remain significant after this correction; the reported p-values are the Holm-adjusted values.

TABLE 5. Performance comparison on the test set (3,600 QA pairs). All MoLE improvements over Single-LoRA are statistically significant ($p < 0.01$, paired bootstrap test with 1,000 resamples).

Metric	Base	LLaVA-Med	S-LoRA	S-LoRA (3×r)	MoLE
BLEU-1	.168	.191	.234	.241	**.279**
BLEU-4	.022	.031	.059	.063	**.075**
ROUGE-L	.177	.208	.251	.258	**.293**
METEOR	.162	.198	.272	.281	**.318**
CIDEr	.091	.146	.318	.334	**.379**
Binary Acc. (%)	22.1	38.6	66.7	68.2	**79.4**
Med. Terms/Ans	0.71	1.94	3.82	3.91	**4.53**

MoLE outperforms the Single-LoRA baseline across all metrics, with improvements of 16.9% in METEOR (0.272 → 0.318), 19.1% in CIDEr (0.318 → 0.379), and 27.1% in BLEU-4 (0.059 → 0.075). All differences are statistically significant at $p < 0.01$ or below. Binary accuracy on yes/no questions improves from 66.7% to 79.4% ($p < 0.001$).

Importantly, MoLE also outperforms the Single-LoRA (3×r) baseline, which matches MoLE's total parameter budget. This demonstrates that the performance gains arise from the mixture-of-experts routing mechanism rather than from increased parameter capacity alone.

LLaVA-Med, despite its biomedical pretraining, underperforms the domain-specific Single-LoRA baseline, indicating that targeted fine-tuning on dermoscopic data is more effective for this specialized task than general medical pretraining alone.

Figure 6 compares the frequency of commonly used dermoscopic terms across models. MoLE's answers contain substantially more clinical vocabulary than the base model and approach reference-answer term frequencies more closely than the Single-LoRA baseline.

F. ADDITIONAL CONTROLLED BASELINES

To isolate the source of MoLE's gains more precisely, we evaluated two further controls on the same test split (Table 6). The first, **LLaVA-Med + LoRA**, fine-tunes the biomedical-pretrained LLaVA-Med backbone with a single dermoscopic LoRA adapter under our exact training protocol; this provides a training-symmetric comparison rather than the zero-shot LLaVA-Med transfer reported in Table 5. The second, the **All-expert ensemble**, retains the three trained LoRA experts but discards the learned router, instead averaging all three expert outputs with equal weight.

TABLE 6. Additional controlled baselines on the test set. Single-LoRA and MoLE are repeated from Table 5 for reference.

Model	BLEU-4	METEOR	CIDEr	Bin. Acc. (%)	Lat. (s)
Single-LoRA	.059	.272	.318	66.7	2.18
LLaVA-Med + LoRA	.061	.286	.342	70.4	2.23
All-expert ensemble	.072	.302	.365	74.2	6.24
MoLE	**.075**	**.318**	**.379**	**79.4**	2.17

Two observations follow. First, training-symmetric fine-tuning of LLaVA-Med (METEOR 0.286, binary accuracy 70.4%) closes much of the gap left by its zero-shot variant but still trails MoLE by 11.2% in METEOR; biomedical pretraining therefore helps when matched for training, yet does not substitute for the heterogeneous specialization MoLE provides. Second, the router-free all-expert ensemble (METEOR 0.302, binary accuracy 74.2%) outperforms a single adapter, confirming that multiple LoRA experts add capacity, but it remains below MoLE (METEOR 0.318, binary accuracy 79.4%) while incurring roughly 3× the inference latency (6.24 s vs. 2.17 s) because every expert must be evaluated for each input. Learned top-1 routing thus matches or exceeds the ensemble's accuracy at one-third of its inference cost, isolating the benefit of routing from that of additional adapter capacity. MoLE's METEOR interval (0.304–0.332) is disjoint from LLaVA-Med + LoRA's (0.272–0.300) and overlaps the ensemble's (0.288–0.316) only marginally, so its advantage over the ensemble is primarily one of efficiency.

G. PER-QUESTION-TYPE PERFORMANCE

Table 7 breaks down METEOR scores by question type for MoLE and the Single-LoRA baseline. MoLE achieves consistent improvements across all categories, with the largest gains on diagnostic questions (+21.4%) and feature-detection questions (+18.6%), precisely the categories where expert specialization is most pronounced.

TABLE 7. METEOR scores by question type. Δ denotes relative improvement of MoLE over Single-LoRA.

Question Type	Single-LoRA	MoLE	Δ
Diagnosis	0.243	0.295	+21.4%
Feature Detection	0.269	0.319	+18.6%
Visual Observation	0.281	0.322	+14.6%
Pattern Description	0.258	0.301	+16.7%
Body Site	0.287	0.324	+12.9%
General Description	0.274	0.312	+13.9%
Yes/No	0.266	0.314	+18.0%

H. ABLATION STUDIES

1) NUMBER OF EXPERTS

Table 8 shows how the performance changes with an increasing number of experts (K). Using a single expert ($K = 1$) results in similar performance to standard

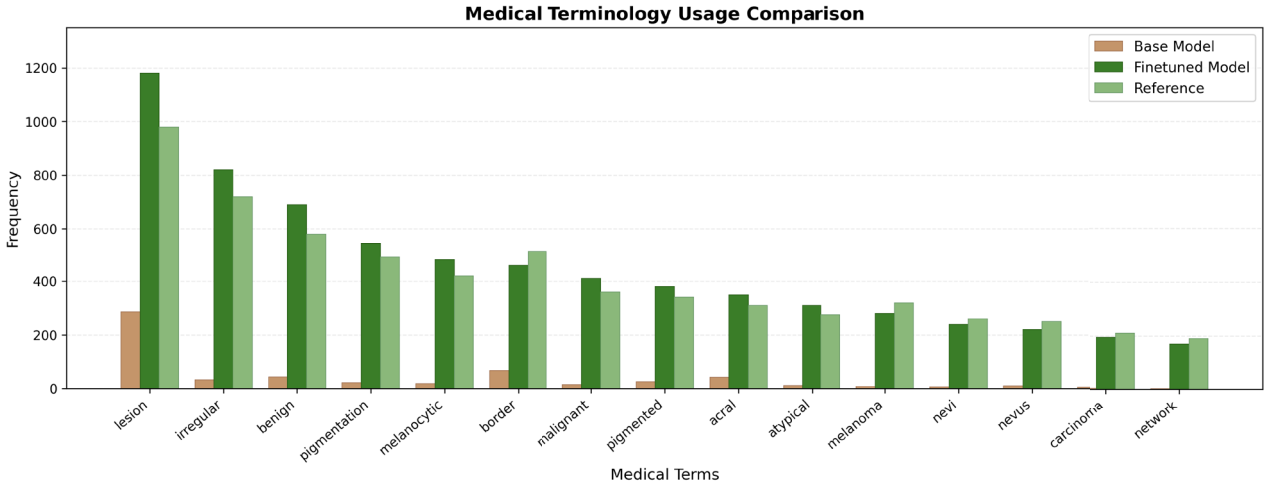


FIGURE 6. Frequency of top medical terms in generated answers. MoLE generates substantially more dermoscopic terminology than the base model and uses terms at frequencies closer to the reference answers.

TABLE 8. Ablation on the number of experts (K). All models use rank $r = 16$.

K	BLEU-4	METEOR	ROUGE-L	Binary Acc.
1	0.059	0.272	0.251	66.7%
2	0.066	0.296	0.271	73.8%
3	**0.075**	**0.318**	**0.293**	**79.4%**
5	0.069	0.305	0.280	76.1%

TABLE 10. Ablation on routing strategy. All models use $K = 3$ experts with $r = 16$.

Routing Strategy	BLEU-4	METEOR	ROUGE-L	Binary Acc.
Random assignment	0.056	0.264	0.248	64.3%
Round-robin	0.058	0.271	0.253	66.1%
Soft routing (avg.)	0.072	0.311	0.286	77.2%
Learned top-1 (ours)	**0.075**	**0.318**	**0.293**	**79.4%**

Single-LoRA. Adding a second expert improves it and $K = 3$ achieves the best overall performance. Increasing experts to $K = 5$ slightly decreases the performance. This is due to over specialization within the training set that is too small to support five distinct specializations.

2) LORA RANK

Table 9 shows the effect of LoRA rank with $K = 3$ experts. Rank $r = 16$ provides the best balance between expressive capacity and regularization. Lower ranks ($r = 4$ and $r = 8$) reduce capacity, whereas $r = 32$ shows evidence of overfitting to the training set.

TABLE 9. Effect of LoRA rank (r) on performance with $K = 3$ experts.

r	BLEU-4	METEOR	ROUGE-L	Binary Acc.
4	0.058	0.281	0.261	71.2%
8	0.066	0.298	0.275	75.6%
16	**0.075**	**0.318**	**0.293**	**79.4%**
32	0.071	0.309	0.284	77.8%

3) ROUTING STRATEGY

Table 10 compares the learned top-1 router against three alternative routing strategies, all using $K = 3$ experts with $r = 16$.

Random and round-robin assignment perform comparably to a single expert, confirming that without learned routing, multiple adapters provide little benefit. Soft routing, which computes a weighted average of expert outputs, performs well but slightly below learned top-1 routing, likely because averaging dilutes the specialized representations. The learned top-1 strategy achieves the best results while maintaining the computational efficiency of single-expert inference.

4) VISION-ENCODER FREEZING

Our default configuration freezes both the CLIP vision encoder and the MLP projector, adapting only the language model. Table 11 examines whether unfreezing visual components helps. Training the projector raises peak memory from 38.4 GB to 41.7 GB and does not improve binary accuracy (it falls from 79.4% to 75.8% with an overlapping METEOR interval), while unfreezing the final four vision-transformer blocks raises peak memory to 47.9 GB—near the 48 GB device limit—and lowers METEOR to 0.296. Because the ISIC training subset is small, adapting the high-capacity visual backbone tends to overfit; freezing it is both more memory-efficient and more accurate, justifying the design choice in Section IV-E.

5) ROUTER DESIGN

Table 12 compares router architectures, all with $K = 3$ experts and $r = 16$. A linear router is clearly

TABLE 11. Vision-encoder freezing ablation ($K = 3$, $r = 16$).

Configuration	METEOR	ROUGE-L	Bin. Acc.	VRAM (GB)
Frozen encoder + frozen projector (ours)	**0.318**	**0.293**	**79.4%**	38.4
Frozen encoder + trained projector	0.312	0.291	75.8%	41.7
Unfreeze final 4 vision blocks	0.296	0.282	77.2%	47.9

weaker (METEOR 0.301, binary accuracy 73.3%) and uses the experts less evenly. Performance improves with router depth and saturates at the two-hidden-layer design we adopt (METEOR 0.318, binary accuracy 79.4%); an attention-pooled router gives comparable METEOR at higher overhead. Top-2 routing attains the best BLEU-4 and the most balanced expert usage but does not improve METEOR or binary accuracy, and it activates two experts per token, increasing compute—so we retain top-1 routing with the two-hidden-layer router.

TABLE 12. Router-design ablation ($K = 3$, $r = 16$, top-1 unless noted).

Router	BLEU-4	METEOR	ROUGE-L	Bin. Acc.
Linear	0.067	0.301	0.276	73.3%
One hidden layer	0.072	0.307	0.286	76.1%
Two hidden layers (ours)	0.075	**0.318**	**0.293**	**79.4%**
Attention-pooled	0.073	0.306	0.292	75.4%
Top-2 routing	**0.076**	0.313	0.287	75.2%

6) EXPERT COLLAPSE AND UTILIZATION

Beyond the load-balancing loss value reported during training, we characterize expert utilization directly using two measures computed over test-set routing: the effective number of experts (the perplexity of the usage distribution, maximal at 3.0 for uniform usage) and the usage Gini coefficient (0 for perfectly balanced usage). Table 13 reports these alongside per-expert usage shares for the full model and for the two auxiliary-loss ablations. Removing the load-balancing loss causes clear routing collapse: usage concentrates on Expert 0 (68.4%), the effective expert count drops to 2.08, the Gini coefficient rises to 0.408, and performance falls sharply (METEOR 0.274, binary accuracy 64.1%)—to roughly the single-adapter level. Removing only the router z-loss is less severe but still imbalanced (effective experts 2.52, Gini 0.239). The full objective maintains balanced utilization (effective experts 2.91, Gini 0.125) and the best performance, confirming that both auxiliary losses are necessary to prevent collapse and that the observed specialization coexists with—rather than contradicts—balanced expert usage.

I. ERROR ANALYSIS BY LESION CATEGORY

We conducted a systematic error analysis on a stratified random sample of 500 test-set responses, annotated by two trained evaluators using predefined error categories. Responses were sampled proportionally to lesion category: 350 benign, 120 malignant, and 30 indeterminate, reflecting the diagnostic distribution of the test set.

TABLE 13. Expert utilization and collapse analysis on the test set.

Configuration	Usage	Eff. exp.	Gini	METEOR	Bin. Acc.
Full objective (ours)	.42/.35/.23	2.91	0.125	**0.318**	**79.4%**
No router z-loss	.53/.30/.17	2.52	0.239	0.300	74.5%
No load-balance loss	.68/.24/.07	2.08	0.408	0.274	64.1%

TABLE 14. Overall error analysis on 500 sampled MoLE responses. Categories are mutually exclusive; a response is assigned to its most severe applicable error type.

Error Category	Count	Rate
Correct and clinically appropriate	341	68.2%
Minor terminology imprecision	72	14.4%
Incomplete but not incorrect	43	8.6%
Hallucinated features not present in image	24	4.8%
Incorrect diagnostic suggestion	14	2.8%
Unsafe or misleading recommendation	6	1.2%

TABLE 15. Error analysis stratified by lesion category (benign $n = 350$; malignant $n = 120$; indeterminate $n = 30$). Numbers are counts; percentages in parentheses.

Error Category	Benign	Malignant	Indeterminate
Correct and clinically appropriate	255 (72.9%)	73 (60.8%)	13 (43.3%)
Minor terminology imprecision	49 (14.0%)	19 (15.8%)	4 (13.3%)
Incomplete but not incorrect	28 (8.0%)	15 (12.5%)	0 (0.0%)
Hallucinated features	11 (3.1%)	10 (8.3%)	3 (10.0%)
Incorrect diagnostic suggestion	5 (1.4%)	7 (5.8%)	2 (6.7%)
Unsafe or misleading	2 (0.6%)	3 (2.5%)	2 (6.7%)

The error rate varies based on lesion category as seen in Tables 14, and 15. For benign lesions, there were a total of 72.9% of fully correct answers, and 0.6% unsafe recommendations. For malignant lesions, the performance decreased to 60.8% correct answers, 8.3% hallucinated features, and 2.5% unsafe answers that could be considered dangerous (e.g., did not include a recommendation for biopsy). As expected clinically, indeterminate lesions had the worst performance with 43.3% correct answers, and 6.7% unsafe answers. Indeterminate lesions such as solar keratoses lack canonical dermoscopic features, making them the most challenging category for any system.

Malignant and indeterminate cases had the highest hallucination rates, at 8.3% and 10.0%, respectively, compared with 3.1% for benign lesions. These categories frequently contain atypical patterns that are underrepresented in the training data and are therefore more likely to elicit unsupported feature combinations. On the same 500-response evaluation set, MoLE reduced hallucinated features by approximately 36% and unsafe omissions by approximately 50% relative to Single-LoRA. Within the malignant category, the corresponding reductions were approximately 29% and 40%.

J. STRUCTURED CLINICAL METRICS

Because linguistic-similarity metrics (BLEU, METEOR, CIDEr) measure overlap with reference text rather than medical correctness, we additionally report structured clinical metrics computed from per-example model outputs against the clinician-reviewed feature labels. These quantify (i) dermoscopic feature-level accuracy and (ii) malignant-lesion detection sensitivity, which we report separately because malignant cases are the highest-risk category.

1) DERMOSCOPIC FEATURE-LEVEL ACCURACY

For every response we compared the model's assertions about individual dermoscopic findings (e.g., pigment network, globules, streaks, blue-white veil, vascular structures) against the reference feature labels, pooling 2,656 feature labels across the stratified review sample. Table 16 reports micro-averaged feature accuracy for the four trained models. MoLE attains the highest feature accuracy (0.808), ahead of the router-free all-expert ensemble (0.790), training-symmetric LLaVA-Med + LoRA (0.753), and Single-LoRA (0.739). Stratified by lesion group, MoLE feature accuracy is 0.831 (95% CI 0.792–0.870) on benign lesions, 0.750 (0.672–0.828) on malignant lesions, and 0.667 (0.498–0.835) on the small indeterminate stratum, mirroring the category difficulty seen in the error analysis (Table 15).

TABLE 16. Micro-averaged dermoscopic feature-level accuracy by model.

Model	Feature Acc.	95% CI
Single-LoRA	0.739	0.722–0.756
LLaVA-Med + LoRA	0.753	0.737–0.769
All-expert ensemble	0.790	0.775–0.805
MoLE	**0.808**	0.793–0.823

2) MALIGNANT-LESION SENSITIVITY

We further isolated the 342 diagnostic-question responses (82 concerning malignant lesions, 260 concerning benign or indeterminate lesions) and scored each model on whether it correctly identified malignancy. Table 17 reports sensitivity (true-positive rate for malignant lesions) and specificity. MoLE achieves the highest malignant sensitivity, 0.793 (65/82; 95% CI 0.705–0.881), compared with 0.756 for the all-expert ensemble, 0.683 for LLaVA-Med + LoRA, and 0.646 for Single-LoRA, while maintaining comparable specificity (≈0.85) across all models. Even so, a malignant sensitivity below 0.80 means roughly one in five malignant diagnostic questions is not flagged as malignant by the model alone, reinforcing the need for the malignant-lesion safeguards described in Section IV-K.

These structured metrics complement, rather than replace, the n-gram metrics: they confirm that MoLE's linguistic improvements are accompanied by gains in feature-level correctness and malignant-lesion sensitivity, while making explicit that absolute safety on high-risk lesions is not yet sufficient for unsupervised use.

TABLE 17. Malignant-lesion detection on the diagnostic-question subset.

Model	Sensitivity	95% CI	Specificity	95% CI
Single-LoRA	0.646	0.543–0.750	0.842	0.798–0.887
LLaVA-Med + LoRA	0.683	0.582–0.784	0.850	0.807–0.893
All-expert ensemble	0.756	0.663–0.849	0.835	0.790–0.880
MoLE	**0.793**	0.705–0.881	**0.854**	0.811–0.897

K. SAFETY AND UNCERTAINTY ANALYSIS

Clinical deployment of VQA systems also creates unique safety concerns beyond the use of typical accuracy measures. Therefore we have carried out a focused examination of responses that could be considered unsafe and analyzed the model's routing confidence as a lightweight, empirically motivated—but, as the calibration analysis below shows, imperfect—signal of output reliability, rather than as a calibrated measure of clinical or epistemic uncertainty.

1) UNSAFE RESPONSE CHARACTERIZATION

The six unsafe responses in the MoLE sample (see Table 14) were evaluated independently by the same two dermatologists. Each response omitted clinically important information rather than providing an incorrect recommendation. Four responses described lesions with malignant features but did not recommend further evaluation or biopsy. Two described pigmented lesions on acral sites without including acral lentiginous melanoma in the differential diagnosis or recommending specialist assessment. No response contained a directly harmful recommendation, such as classifying a malignant lesion as benign. All six responses were routed to the diagnostic reasoning expert (Expert 0), indicating that safety-critical omissions were concentrated among clinically difficult questions.

For comparison, the Single-LoRA baseline produced 12 unsafe responses (2.4%) on the same 500-sample set, of which five involved active misclassification of malignant lesions as benign-appearing. MoLE's unsafe responses were qualitatively less severe.

2) ROUTING CONFIDENCE AS AN UNCERTAINTY SIGNAL

We examined whether low routing confidence (high routing entropy) predicts higher error rates. Responses were binned by the maximum routing probability $p_{\max}$ of the assigned expert into three groups: high confidence ($p_{\max} > 0.8$; $n = 207$), moderate confidence ($0.5 < p_{\max} \leq 0.8$; $n = 220$), and low confidence ($p_{\max} \leq 0.5$; $n = 73$). Error rates within each bin are reported in Table 18.

TABLE 18. Error rates by routing confidence bin on the 500-sample error analysis set.

$p_{\max}$ bin	n	Correct (%)	Hallucination (%)	Unsafe (%)
High (> 0.8)	207	74.9	3.4	0.5
Moderate (0.5–0.8)	220	66.8	4.5	1.4
Low (≤ 0.5)	73	52.1	9.6	2.7

Routing confidence has a monotonic association with response quality: high-confidence assignments are correct in 74.9% of cases with 0.5% unsafe omissions, whereas low-confidence assignments are correct in 52.1% of cases with 2.7% unsafe omissions. A threshold of $p_{\max} \leq 0.5$ can therefore serve as a conservative expert-selection flag for human review or an explicit caution notice, but it must not be interpreted as a calibrated estimate of factual reliability. Applying this threshold to the test set would flag 73 of 3,600 responses (2.0%). Low-confidence cases are concentrated in the indeterminate category (34% of indeterminate responses, compared with 14% of benign and 18% of malignant responses), consistent with the ambiguity of questions about visually uncertain lesions.

3) CALIBRATION ANALYSIS

The monotonic relationship in Table 18 establishes that routing confidence correlates with correctness, but it does not establish that routing confidence is a *calibrated* or *reliable* detector of unsafe outputs. To test this directly, we paired the per-example routing probabilities with correctness and safety labels on the 500-sample error-analysis set and computed two complementary artifacts: a reliability diagram (Figure 7) and threshold-free detection metrics (AUROC and AUPRC) using $1 - p_{\max}$ as the alarm score (Table 19).

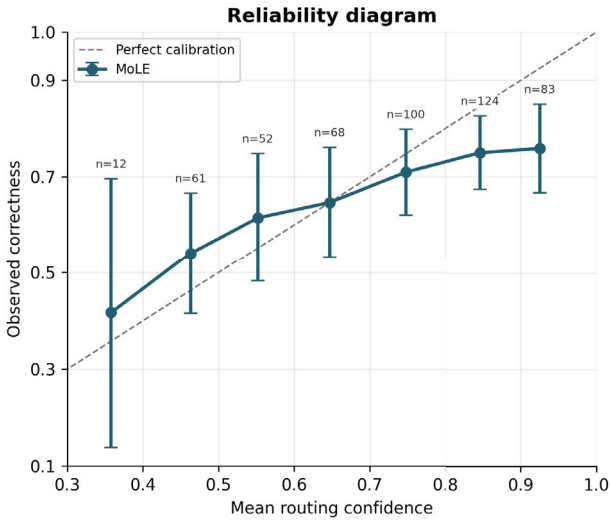


FIGURE 7. Reliability diagram of routing confidence versus observed correctness (500-sample set).

The reliability diagram shows that observed correctness rises monotonically with routing confidence—from 0.42 in the lowest populated bin to 0.76 in the highest—confirming the qualitative trend. However, the curve plateaus well below the perfect-calibration diagonal in the high-confidence region: even the most confident responses ($p_{\max} > 0.9$) are correct only ∼76% of the time. Routing confidence is therefore a useful but poorly calibrated signal; it systematically over-estimates reliability at the high end.

TABLE 19. Detection of incorrect and unsafe responses using $1 - p_{\max}$ as the alarm score.

Target	n	AUROC	AUPRC
Any incorrect response	159	0.594 (0.541–0.647)	0.381 (0.323–0.443)
Hallucinated feature	24	0.623 (0.506–0.738)	0.086 (0.041–0.151)
Unsafe omission	6	0.548 (0.361–0.724)	0.019 (0.008–0.052)
Hallucinated or unsafe	28	0.618 (0.507–0.724)	0.112 (0.055–0.180)

The detection metrics quantify this limitation. Routing confidence detects *any* incorrect response only modestly above chance (AUROC 0.594) and detects the clinically critical categories scarcely better: AUROC 0.623 for hallucinated features and 0.548—essentially chance, with a confidence interval spanning 0.5—for unsafe omissions. The corresponding AUPRC values are very low for the rare safety targets (0.086 for hallucination, 0.019 for unsafe omission), reflecting both their low base rate and the weakness of the signal. In other words, routing confidence cannot be relied upon to catch high-confidence unsafe responses.

TABLE 20. Representative high-confidence ($p_{\max} > 0.83$) responses that were incorrect or unsafe.

Lesion	$p_{\max}$	Expert	Issue
Malignant	0.914	0	Unsafe omission: described asymmetry and border irregularity but gave no specialist/biopsy recommendation.
Malignant	0.889	1	Hallucinated feature: asserted a blue-white veil that was absent on clinical review.
Indeterminate	0.862	0	Incorrect suggestion: leaned toward benign nevus despite keratotic and atypical cues.
Malignant	0.846	1	Incorrect feature: denied an irregular pigment network marked present.
Benign	0.831	2	Hallucinated feature: added vascular structures absent from the reference review.

Table 20 illustrates this failure mode: each listed response had routing confidence above 0.83 but was incorrect or unsafe, and three of the five involved malignant lesions. A high routing probability reflects confident *expert selection*, not clinical reliability. The low-confidence flag ($p_{\max} \leq 0.5$) can support escalation only when combined with content-based safeguards and mandatory clinical review.

A more robust deployment fallback should combine the routing flag with a deterministic lexical safety filter. The filter would scan both the question and generated answer for high-risk concepts such as *melanoma*, *malignant*, *biopsy*, *atypical*, and *ulceration*. When a high-risk concept is present, the interface should display a standardized non-diagnostic disclaimer and recommend specialist assessment regardless of routing confidence. It should also detect the absence of referral or biopsy language when the generated answer raises a malignant differential. This rule-based layer is proposed as

a transparent fail-safe and was not implemented or clinically validated in the present prototype.

4) SCOPE AND DISCLAIMER

These findings reinforce the position stated in the Introduction: MoLE must not be used as a standalone diagnostic tool. Outputs for malignant and indeterminate lesions have a non-negligible rate of incompleteness or unsafe omission. The appropriate context is an educational or second-opinion aid under the supervision of a qualified dermatologist, with mandatory human review for low-routing-confidence responses and for questions involving potentially malignant lesions.

L. CLINICIAN EVALUATION

Three board-certified dermatologists independently evaluated a blinded set of 200 randomly sampled test-set question–image pairs. For each pair, evaluators received the reference answer and shuffled responses from three systems (Base, Single-LoRA, MoLE) without model labels. Each response was rated on a 1–5 Likert scale across three dimensions: *Clinical Accuracy* (correctness of clinical assertions and diagnostic reasoning), *Terminology Appropriateness* (use of standard dermoscopic vocabulary), and *Completeness* (coverage of relevant findings for the given question).

TABLE 21. Clinician evaluation (mean ± SD, three evaluators). All MoLE improvements over Single-LoRA are statistically significant ($p < 0.01$, paired t-test, $n = 200$).

Dimension	Base	Single-LoRA	MoLE
Clinical Accuracy	1.82 ± 0.74	3.14 ± 0.68	$\mathbf{3.71 \pm 0.61}$
Terminology	1.64 ± 0.81	3.28 ± 0.72	$\mathbf{3.86 \pm 0.58}$
Completeness	2.01 ± 0.69	3.06 ± 0.65	$\mathbf{3.52 \pm 0.59}$
Overall	1.82 ± 0.62	3.16 ± 0.54	$\mathbf{3.70 \pm 0.48}$

Table 21 provides an overview of the findings. MoLE is statistically better over Single-LoRA on all three dimensions ($p < 0.01$ or below, paired t-test). The level of inter-rater reliability, as indicated by Krippendorff's α at.76, demonstrated an acceptable level of agreement among evaluators. Also, it is evident from the results that the greatest improvements are noted in terminology appropriateness ($3.28 \rightarrow 3.86$).

During the blinded evaluation, dermatologists were also asked to flag any response they considered potentially unsafe for clinical use. Across the 200 evaluated pairs, evaluators flagged 4 MoLE responses (2.0%) and 11 Single-LoRA responses (5.5%) as potentially unsafe, consistent with the quantitative error analysis. All flagged responses involved either omission of biopsy recommendation or failure to note atypical features in high-risk lesions.

M. NEAR-DOMAIN CROSS-DATASET EVALUATION

To assess near-domain robustness across a related dermoscopic source, we evaluated all models on an independent set of 200 images drawn from HAM10000 [35]. Because HAM10000 and the ISIC Archive share contributing institutions—and HAM10000 images are registered in the ISIC Archive under their own identifiers—we performed explicit image-level de-duplication before sampling. First, any HAM10000 image whose ISIC identifier matched an image in our training or test pool was excluded. Second, to detect re-scans, crops, or re-compressions that share content but not identifiers, we screened the remaining candidates against the complete ISIC pool using perceptual hashing (pHash) and cosine similarity between CLIP image embeddings, discarding candidates above a conservative similarity threshold. The 200 retained images are therefore disjoint at the image level from the ISIC data used for training and testing. They represent different acquisition devices and clinical sites and have a different class balance, but remain within the dermoscopic modality and share sourcing with ISIC. HAM10000 follows a seven-class taxonomy comprising actinic keratoses/intraepithelial carcinoma, basal cell carcinoma, benign keratosis-like lesions, dermatofibroma, melanoma, melanocytic nevi, and vascular lesions. Following the protocol in Section III-B2, we generated 20 QA pairs per image (4,000 pairs total) using the same question schema; one board-certified dermatologist reviewed these pairs for clinical accuracy.

TABLE 22. Near-domain cross-dataset evaluation on 200 HAM10000 images (4,000 QA pairs). Results are lower than on the in-distribution test set for all models, while their relative ordering is preserved.

Model	BLEU-4	METEOR	ROUGE-L	Binary Acc.
Base LLaVA	0.018	0.142	0.161	19.8%
LLaVA-Med	0.026	0.181	0.194	34.2%
Single-LoRA	0.051	0.241	0.229	59.8%
S-LoRA (3×r)	0.053	0.249	0.237	61.3%
MoLE	**0.065**	**0.284**	**0.271**	**72.1%**

Table 22 shows that all models perform below their in-distribution results, consistent with a shift in imaging devices, patient populations, and lesion subtypes. MoLE retains its relative advantage across all metrics: its HAM10000 METEOR score of 0.284 is 17.8% higher than the Single-LoRA score of 0.241, comparable to the 16.9% relative improvement on the in-distribution test set. The binary-accuracy gap is also similar (72.1% vs. 59.8% on HAM10000 and 79.4% vs. 66.7% on the ISIC test set). These results support near-domain robustness across related dermoscopic sources; they do not establish broad out-of-distribution generalization.

The absolute reduction from the in-distribution test set to HAM10000 is slightly smaller for MoLE (METEOR: $0.318 \rightarrow 0.284$, -10.7%) than for Single-LoRA ($0.272 \rightarrow 0.241$, -11.4%), providing limited evidence of improved robustness to this near-domain shift.

N. EXPERT-SPECIFIC PERFORMANCE

Table 23 summarizes expert-specific performance. Expert 0 handles the largest share of examples (41.8%) and has the

highest METEOR and BLEU-4 scores. Experts 1 and 2, associated with feature detection and general observation, respectively, have lower scores, reflecting differences in the routed question types.

TABLE 23. Performance metrics for each expert on its assigned test-set examples.

Metric	Expert 0	Expert 1	Expert 2	Overall
Examples	1,505 (41.8%)	1,264 (35.1%)	831 (23.1%)	3,600
BLEU-4	0.081	0.072	0.066	0.075
METEOR	0.328	0.314	0.302	0.318
Avg. Loss	0.712	0.738	0.761	0.736
Med. Terms/Ans	4.82	4.47	4.08	4.53

O. QUALITATIVE EXAMPLES

Table 24 presents representative responses from each expert, illustrating the distinct reasoning styles that emerge from the MoLE architecture.

Expert 0 provides diagnostic reasoning based on multiple dermoscopic findings and expresses an appropriate degree of uncertainty. Expert 1 focuses on individual morphologic structures and uses precise dermoscopic terminology. Expert 2 provides systematic visual descriptions without an interpretive diagnosis. These response styles are consistent with the specialization patterns observed in the quantitative routing analysis.

P. DEPLOYMENT FEASIBILITY

To evaluate practical deployment feasibility, we hosted MoLE on a six-node GPU cluster at Namal University. Each node is equipped with an NVIDIA RTX 4070 Ti (12 GB VRAM), and the cluster is managed by SLURM. A Chainlit-based web interface allows users to upload dermoscopic images and ask free-form questions. Model weights are stored in a Lustre distributed file system accessible to all compute nodes. The software environment uses PyTorch 2.5.1 with CUDA 12.2.

With batch size one and greedy decoding, average end-to-end inference latency, including routing, is approximately 2.1 s per question on a single RTX 4070 Ti. Average generation throughput under the same configuration is approximately 28 tokens/s, and routing overhead is less than 5 ms. Figure 8 shows example interactions. The interface displays routing confidence and flags outputs with $p_{\max} \leq 0.5$ for human review. This deployment demonstrates computational feasibility on consumer-grade hardware, not clinical readiness. The interface is a research and educational testbed rather than a patient-facing medical device. Clinical use would require prospective validation, regulatory clearance, and integration into a supervised workflow. Content-based safety filters, including the high-risk keyword fallback described in Section IV-K, should be implemented and validated before any clinical evaluation.

V. DISCUSSION

A. KEY FINDINGS

Our results demonstrate that combining mixture-of-experts routing with parameter-efficient fine-tuning improves performance in specialized medical VQA. Four findings stand out. First, expert specialization emerges from load-balanced training without task labels, with the router partitioning inputs according to the required reasoning mode. Second, the gains are attributable to routing rather than parameter count alone, as shown by MoLE outperforming the parameter-matched Single-LoRA (3×r) baseline. Third, automatic metrics and blinded clinician evaluation provide convergent evidence of improved response quality. Fourth, the relative performance advantage persists in the near-domain HAM10000 evaluation. Because HAM10000 shares the dermoscopic modality and sourcing with ISIC, this result should be interpreted as robustness to a related source rather than broad clinical generalization.

B. COMPARISON WITH RELATED WORK

MoLE extends the mixture-of-LoRA paradigm presented by Wu et al. [12], which combines multiple LoRA adapters across NLP and vision–language settings, to intra-domain reasoning specialization in dermoscopic VQA. In our system, the experts are learned jointly within a single clinical domain and specialize according to the cognitive demands of each question type.

Compared with medical VLMs such as LLaVA-Med [7], which apply uniform fine-tuning across all medical tasks, MoLE achieves stronger performance on dermoscopic VQA through domain-specific specialization. The zero-shot performance of LLaVA-Med on our test set (Table 5) confirms that general biomedical pretraining is insufficient for specialized dermoscopic interpretation and that targeted adaptation is essential.

C. ANALYSIS OF EXPERT BEHAVIOR

Expert 0 (diagnostic reasoning) uses a higher number of disease names, diagnostic qualifiers, and hedging language (i.e., ‘‘suggestive of,’’ ‘‘warrants consideration’’). Expert 1 (feature detection) uses a greater number of morphological descriptors (e.g., ‘‘reticular network’’, ‘‘globular pattern’’, ‘‘arborizing vessels’’). Expert 2 (general observation) provides systematic visual description, using fewer diagnostic commitments. All these linguistic specializations were developed in the absence of explicit supervision.

The routing-confidence analysis shows that 85.4% of inputs receive a clear expert preference ($p > 0.5$), while 14.6% exhibit ambiguous routing. Ambiguous cases often involve questions that span multiple reasoning modes and may benefit from soft routing or expert ensembling. Although low-confidence assignments have higher error rates (Table 18), routing probability measures expert-selection confidence and is not a calibrated estimate of answer reliability.

TABLE 24. Example responses demonstrating expert specialization patterns.

Q1: What is the most likely diagnosis? (Routed to Expert 0) **MoLE:** This lesion shows features suggestive of a melanocytic nevus. The relatively symmetrical appearance, regular borders, and homogeneous brown pigmentation are consistent with a benign nevus. However, the presence of some asymmetry and border irregularity warrants consideration of an atypical nevus or early melanoma in the differential diagnosis. Clinical correlation and possible biopsy may be appropriate.
Q2: Is there a pigment network present? (Routed to Expert 1) **MoLE:** Yes, there is a pigment network visible. The network appears as a mesh-like pattern of brown lines with lighter areas in between, characteristic of a reticular dermoscopic pattern. The network is relatively regular with consistent pigmentation, which is a reassuring feature.
Q3: Describe the overall appearance of this lesion. (Routed to Expert 2) **MoLE:** The lesion appears as a light brown, oval-shaped area with well-defined borders. The surface shows relatively uniform pigmentation without obvious color variation. The size appears modest, likely several millimeters in diameter. The surrounding skin appears normal without inflammation or other changes.

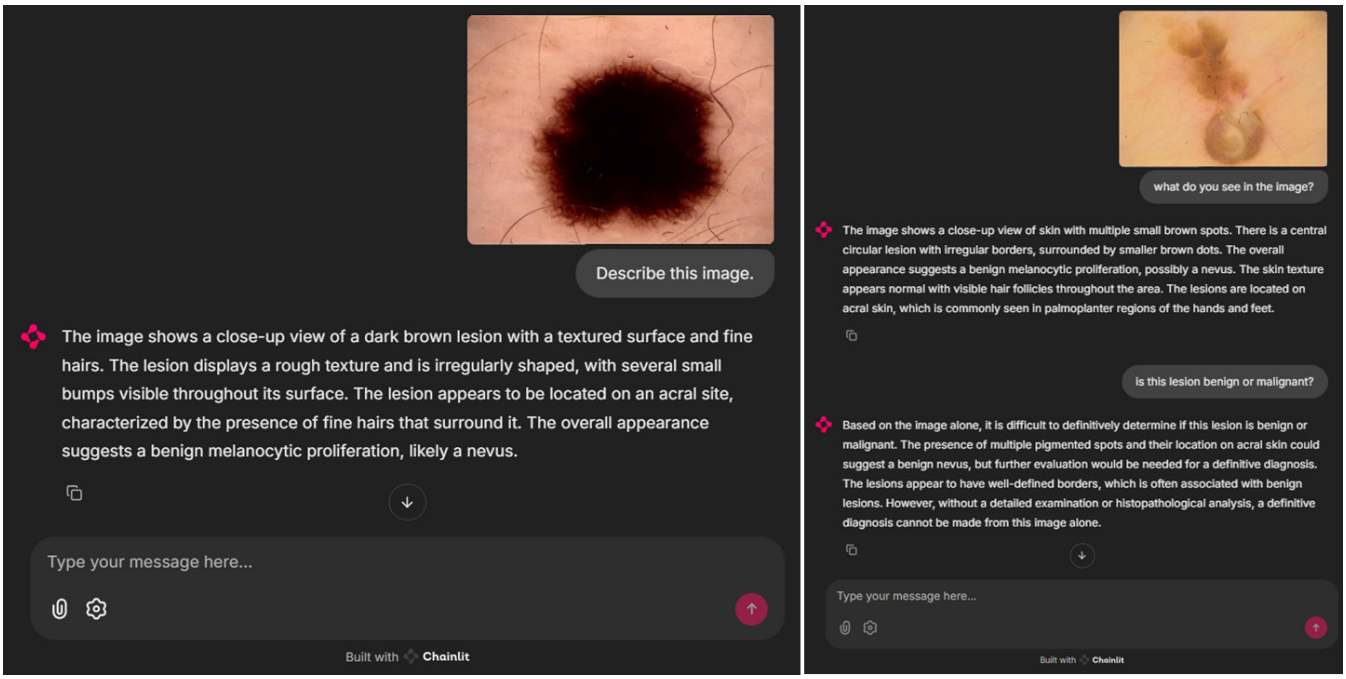


FIGURE 8. Web-based interface for interactive dermoscopic VQA, deployed on the Namal University HPC cluster.

A natural objection is that the observed specialization is merely an artifact of the load-balancing objective rather than genuine emergent behavior. We argue otherwise on three grounds. First, the load-balancing loss in Eq. (12) penalizes *deviation from uniform expert usage*; it therefore pushes the router toward an even split of examples and is agnostic to question type. It cannot, by construction, reward aligning a particular expert with a particular reasoning category—yet such alignment is exactly what we observe in Table 3 (e.g., 71.2% of diagnostic questions route to Expert 0). The auxiliary loss thus works *against*, not toward, the reported specialization. Second, the control baselines isolate the effect: random and round-robin assignment, which also satisfy load balancing, perform no better than a single adapter (Table 10), and the router-free all-expert ensemble—which has the same adapter capacity but no learned partition—also trails MoLE (Table 6), so neither balanced utilization nor added capacity alone produces the gains. Moreover, the utilization analysis (Table 13) shows that balanced usage and type-aligned specialization coexist: the full model is both well balanced (Gini 0.125) and strongly type-aligned, whereas removing the balancing loss collapses usage onto one expert *and* degrades performance, the opposite of what an "artifact of balancing" account would predict. Third, the specialization is consistent across orthogonal measures—routing distribution (Table 3), per-expert loss and terminology counts (Table 23), and qualitative response style (Table 24)—which would not be expected from a routing pattern driven solely by a balancing penalty. We nonetheless agree that the expert *labels* we assign ("diagnostic," "feature detection," "observation")

are post-hoc interpretations of an emergent partition rather than architecturally enforced roles, and we treat them as descriptive throughout.

D. SAFETY CONSIDERATIONS

The stratified error analysis (Section IV-I) and safety analysis (Section IV-K) show that unsafe omissions are concentrated in malignant and indeterminate lesions. A deployment system should therefore apply content-based safeguards so that outputs mentioning malignant or otherwise high-risk findings are accompanied by a standardized disclaimer and recommendation for specialist assessment. A transparent rule-based checker for high-risk concepts can provide this fallback regardless of routing confidence. The $p_{\max} \leq 0.5$ threshold can serve as an additional escalation flag, but it should not be used as a safety assurance.

E. METHODOLOGICAL GENERALIZABILITY BEYOND DERMOSCOPY

Although our experiments are confined to dermoscopic VQA, the MoLE formulation is largely modality-agnostic: it requires only a frozen backbone, a small set of low-rank experts, and a learned router over a pooled input representation. This suggests applicability well beyond dermatology, and indeed beyond images. In particular, the same mechanism could be applied to non-image biomedical signals—such as electroencephalography (EEG), respiratory sounds, or other physiological time series—where the heterogeneity that MoLE exploits arises naturally. In those settings, the latent axis of expert specialization need not be visual reasoning category; it could instead correspond to signal modality, temporal or sleep state, task type, or pathological pattern. For example, multi-view and brain-network architectures have been used for neonatal sleep staging [38], and multi-task learning has been used to jointly model sleep staging and disorder detection [39]; a MoLE-style router could allocate experts across such co-occurring sub-tasks or physiological regimes rather than across question types. Similar opportunities exist in physiological-measurement-based screening pipelines, such as those surveyed for chronic obstructive pulmonary disease [40]. We therefore view intra-domain, label-free expert specialization as a general design pattern for heterogeneous biomedical reasoning, of which dermoscopic VQA is one instance; validating this breadth on physiological-signal modalities is a promising direction for future work.

F. LIMITATIONS

Several limitations should be considered. *Dataset scale and source.* The 1,200-image dataset remains small relative to general benchmarks and is drawn from a single archive, which limits diversity in acquisition devices, demographics, and geography. *Strength of the robustness test.* HAM10000 is dermoscopic and shares sourcing with ISIC; the cross-dataset evaluation therefore constitutes a near-domain robustness check rather than a strong out-of-distribution test. Multi-institutional data spanning different devices, populations, and geographic regions are required before making clinical generalization claims. *Annotation provenance.* Trained research engineers produced the primary annotations. Dermatologists reviewed 3,600 pairs (15%) and corrected 461 of them, but the remaining pairs did not receive item-level dermatologist review. *Clinician study scale.* Three dermatologists rated 200 samples in the blinded evaluation; this scale limits the strength and generalizability of the clinical conclusions, and a larger prospective reader study is needed. *Nature of expert roles.* Expert roles emerge during training and are interpreted post hoc rather than enforced architecturally, so their stability across random seeds and datasets is not guaranteed. *Uncertainty estimates.* Routing probability reflects expert-selection confidence rather than calibrated epistemic uncertainty over generated text. The calibration analysis (Section IV-K3) shows that it is poorly calibrated at the high end and is a weak detector of unsafe outputs (AUROC 0.55–0.62 for hallucinated features and unsafe omissions). It cannot reliably detect high-confidence unsafe responses. *Evaluation metrics.* BLEU, ROUGE, METEOR, and CIDEr capture similarity to reference text rather than complete medical factuality. We complement them with structured clinical metrics, stratified error analysis, and clinician ratings, but broader clinical evaluation remains necessary.

G. FUTURE DIRECTIONS

Future evaluation should use multi-institutional data from varied imaging systems and patient populations. A larger clinician study with formal inter-rater analysis and a prospective user study would provide a stronger characterization of utility and failure modes. Technical directions include soft routing for ambiguous inputs, dynamic selection of the number of experts, calibrated text-level uncertainty estimates, and deterministic high-risk keyword and referral checks. Any progression toward clinical decision support would also require prospective safety validation and assessment under the applicable regulatory framework.

VI. REPRODUCIBILITY

To support exact replication, we consolidate here the settings required to reproduce our results. The complete training, inference, and evaluation code, together with the derived QA annotations and split definitions, is released in the project repository (see Code Availability).

Software and hardware. MoLE is implemented in PyTorch 2.5.1 with Hugging Face Transformers 4.36.0 and PEFT. The base LLaVA-1.5-7B model is loaded with 4-bit NF4 quantization (QLoRA). Training was performed on a single NVIDIA L40S GPU (48 GB) and inference was verified on an NVIDIA RTX 4070 Ti (12 GB) running CUDA 12.2.

Determinism. A single fixed random seed governs the image-level train/test split, expert and router initialization,

TABLE 25. Representative question templates for each reasoning type.

Reasoning type	Representative prompt
Diagnosis	"What is the most likely diagnosis for this lesion?"
Feature Detection	"Is a pigment network present in this lesion?"
Visual Observation	"What colours are visible within this lesion?"
Pattern Description	"Which dermoscopic pattern best describes this lesion?"
Body Site	"On which anatomical site is this lesion located?"
General Description	"Describe the overall appearance of this lesion."
Yes/No	"Are blue-white structures present? Answer yes or no."

and batch ordering; the exact seed and configuration files are included in the repository. Because all models are evaluated with greedy decoding, generation is deterministic and the reported metrics are not subject to sampling variance.

Data and preprocessing. Input images are resized to 224×224 before the CLIP ViT-L/14 encoder. The dataset is partitioned by image into 85% training and 15% test (Section III-B2), with all 20 QA pairs for an image confined to a single split. No additional image augmentation is applied.

Input format. Each example follows the LLaVA conversation schema. The user turn is the image placeholder followed by the question, "`<image>\n{question}`", and the target is the reference answer. Representative question templates for the seven reasoning types are listed in Table 25; the full template set and answer-formatting guidelines are in the repository.

Optimization. We train three LoRA experts ($K = 3$, rank $r = 16$, $\alpha = 32$, dropout 0.05) applied to the seven attention and feed-forward projections, plus a router with hidden dimensions 512 and 256 (dropout 0.1). Training uses AdamW for 3 epochs at batch size 32 ($\sim$1,913 steps), with learning rates 5×10^{-5} (experts) and 2.5×10^{-4} (router), a cosine schedule with 300 warmup steps, gradient clipping at norm 1.0, weight decay 0.01 (experts only), and auxiliary-loss weights $\lambda_{balance} = 0.001$ and $\lambda_z = 0.0001$.

Evaluation. Metrics (BLEU, ROUGE-L, METEOR, CIDEr, and binary accuracy) are computed on the held-out test split with greedy decoding. Confidence intervals use 1,000 bootstrap resamples, and significance is assessed with the paired bootstrap test under Holm–Bonferroni correction (Section IV-E).

VII. CONCLUSION

We introduced Mixture-of-LoRA-Experts (MoLE), a parameter-efficient approach to dermoscopic visual question answering that combines multiple LoRA adapters with learned top-1 routing. The model was trained on 1,200 dermoscopic images and 24,000 question–answer pairs and developed distinct response patterns associated with diagnostic reasoning, feature detection, and general observation. MoLE achieved a 16.9% relative improvement in METEOR over Single-LoRA and 79.4% accuracy on binary questions. Ablation studies support the use of three experts, LoRA rank 16, and learned routing. The relative advantage persisted in a near-domain HAM10000 evaluation, although this result does not establish broad out-of-distribution generalization. Error rates and unsafe omissions were highest for malignant and indeterminate lesions. Routing probability was only weakly predictive of unsafe output and should be combined with human review and deterministic content-based safety checks. Training required less than 8 hours on a single 48 GB GPU, and inference achieved approximately 28 tokens/s with 2.1 s average latency on consumer-grade 12 GB hardware. MoLE is a research prototype without clinical validation or regulatory authorization and should not be used as a standalone diagnostic or patient-facing system.

DATA AVAILABILITY

Dermoscopic image data used in this study were obtained from publicly available datasets, including HAM10000 [35] and the ISIC Archive [34], subject to the access terms of those repositories. The derived question–answer annotations, split definitions, and evaluation files generated in this study will be made available through the project GitHub repository: https://github.com/ucerd/llava_skin_cancer

CODE AVAILABILITY

The code required to reproduce model training, inference, and evaluation is available through the project GitHub repository: https://github.com/ucerd/llava_skin_cancer. The repository includes scripts for model fine-tuning, inference, evaluation, and generation of paper-related outputs.

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ETHICS STATEMENT

This study used de-identified dermoscopic images obtained from publicly available archival datasets. No new patient recruitment, intervention, or direct participant contact was involved. The work consisted of secondary analysis of existing public data and associated derived annotations. According to institutional policy, this type of retrospective analysis of publicly available de-identified data did not require formal Institutional Review Board approval.

CONSENT FOR PUBLICATION

This article contains no identifiable personal information requiring consent for publication.

STATEMENT OF APPRECIATION

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COMPETING INTERESTS

The authors declare that they have no competing interests.

REFERENCES

[1] R. L. Siegel, K. D. Miller, N. S. Wagle, and A. Jemal, "Cancer statistics, 2023," *CA: Cancer J. Clinicians*, vol. 73, no. 1, pp. 17–48, Jan. 2023.

[2] *American Cancer Society, Cancer Facts & Figures 2023*. Atlanta, GA, USA: American Cancer Society, 2023.

[3] M. E. Vestergaard, P. Macaskill, P. E. Holt, and S. W. Menzies, "Dermoscopy compared with naked eye examination for the diagnosis of primary melanoma: A meta-analysis of studies performed in a clinical setting," *Brit. J. Dermatology*, vol. 159, no. 3, pp. 669–676, Jun. 2008.

[4] H. A. Haenssle et al., "Man against machine: Diagnostic performance of a deep learning convolutional neural network for dermoscopic melanoma recognition in comparison to 58 dermatologists," *Ann. Oncol.*, vol. 29, no. 8, pp. 1836–1842, 2018.

[5] A. Esteva, B. Kuprel, R. A. Novoa, J. Ko, S. M. Swetter, H. M. Blau, and S. Thrun, "Dermatologist-level classification of skin cancer with deep neural networks," *Nature*, vol. 542, no. 7639, pp. 115–118, Feb. 2017.

[6] H. Liu, C. Li, Q. Wu, and Y. J. Lee, "Visual instruction tuning," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 36, 2023, pp. 34892–34916.

[7] C. Li, C. Wong, S. Zhang, N. Usuyama, H. Liu, J. Yang, T. Naumann, H. Poon, and J. Gao, "LLaVA-Med: Training a large language-and-vision assistant for biomedicine in one day," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 36, 2023, pp. 28541–28564, doi: 10.52202/075280-1240.

[8] E. J. Hu, Y. Shen, P. Wallis, Z. Allen-Zhu, Y. Li, S. Wang, L. Wang, and W. Chen, "LoRA: Low-rank adaptation of large language models," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2022, pp. 1–26.

[9] R. A. Jacobs, M. I. Jordan, S. J. Nowlan, and G. E. Hinton, "Adaptive mixtures of local experts," *Neural Comput.*, vol. 3, no. 1, pp. 79–87, Mar. 1991.

[10] N. Shazeer, A. Mirhoseini, K. Maziarz, A. Davis, Q. Le, G. Hinton, and J. Dean, "Outrageously large neural networks: The sparsely-gated mixture-of-experts layer," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2017, pp. 1–19.

[11] W. Fedus, B. Zoph, and N. Shazeer, "Switch transformers: Scaling to trillion parameter models with simple and efficient sparsity," *J. Mach. Learn. Res.*, vol. 23, no. 120, pp. 1–39, 2022.

[12] X. Wu, S. Huang, and F. Wei, "Mixture of LoRA experts," 2024, *arXiv:2404.13628*, doi: 10.48550/arXiv.2404.13628.

[13] S. Dou, E. Zhou, Y. Liu, S. Gao, W. Shen, L. Xiong, Y. Zhou, X. Wang, Z. Xi, X. Fan, S. Pu, J. Zhu, R. Zheng, T. Gui, Q. Zhang, and X. Huang, "LoRAMoE: Alleviating world knowledge forgetting in large language models via MoE-style plugin," in *Proc. 62nd Annu. Meeting Assoc. Comput. Linguistics*, 2024, pp. 1932–1945.

[14] L. H. Li, M. Yatskar, D. Yin, C.-J. Hsieh, and K.-W. Chang, "VisualBERT: A simple and performant baseline for vision and language," 2019, *arXiv:1908.03557*, doi: 10.48550/arXiv.1908.03557.

[15] H. Tan and M. Bansal, "LXMERT: Learning cross-modality encoder representations from transformers," in *Proc. Conf. Empirical Methods Natural Lang. Process., 9th Int. Joint Conf. Natural Lang. Process. (EMNLP-IJCNLP)*, 2019, pp. 5099–5110.

[16] A. Radford, J. W. Kim, C. Hallacy, A. Ramesh, G. Goh, S. Agarwal, G. Sastry, A. Askell, P. Mishkin, J. Clark, G. Krueger, and I. Sutskever, "Learning transferable visual models from natural language supervision," in *Proc. Int. Conf. Mach. Learn.*, 2021, pp. 8748–8763.

[17] J. Li, D. Li, C. Xiong, and S. Hoi, "BLIP: Bootstrapping language-image pre-training for unified vision-language understanding and generation," in *Proc. Int. Conf. Mach. Learn.*, 2022, pp. 12888–12900.

[18] J. Li, D. Li, S. Savarese, and S. Hoi, "BLIP-2: Bootstrapping language-image pre-training with frozen image encoders and large language models," in *Proc. Int. Conf. Mach. Learn. (ICML)*, vol. 202, 2023, pp. 19730–19742.

[19] H. Liu, C. Li, Y. Li, and Y. J. Lee, "Improved baselines with visual instruction tuning," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2024, pp. 26296–26306.

[20] M. Moor, Q. Huang, S. Wu, M. Yasunaga, Y. Dalmia, J. Leskovec, C. Zakka, E. P. Reis, and P. Rajpurkar, "Med-Flamingo: A multimodal medical few-shot learner," in *Proc. Mach. Learn. Health Symp.*, vol. 225, 2023, pp. 353–367.

[21] C. Wu, X. Zhang, Y. Zhang, H. Hui, Y. Wang, and W. Xie, "Towards generalist foundation model for radiology by leveraging Web-scale 2D&3D medical data," *Nature Commun.*, vol. 16, no. 1, p. 7866, Aug. 2025.

[22] Z. Chen et al., "A vision-language foundation model to enhance efficiency of chest X-ray interpretation," 2024, *arXiv:2401.12208*, doi: 10.48550/arXiv.2401.12208.

[23] M. Y. Lu et al., "A multimodal generative AI copilot for human pathology," *Nature*, vol. 634, no. 8033, pp. 466–473, 2024, doi: 10.1038/s41586-024-07618-3.

[24] S. Zhang et al., "BiomedCLIP: A multimodal biomedical foundation model pretrained from fifteen million scientific image-text pairs," 2023, *arXiv:2303.00915*, doi: 10.48550/arXiv.2303.00915.

[25] T. Dettmers, A. Pagnoni, A. Holtzman, and L. Zettlemoyer, "QLoRA: Efficient finetuning of quantized LLMs," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 36, 2023, pp. 10088–10115, doi: 10.52202/075280-0441.

[26] X. L. Li and P. Liang, "Prefix-tuning: Optimizing continuous prompts for generation," in *Proc. 59th Annu. Meeting Assoc. Comput. Linguistics, 11th Int. Joint Conf. Natural Lang. Process.*, 2021, pp. 4582–4597.

[27] B. Lester, R. Al-Rfou, and N. Constant, "The power of scale for parameter-efficient prompt tuning," in *Proc. Conf. Empirical Methods Natural Lang. Process.*, 2021, pp. 3045–3059.

[28] N. Houlsby, A. Giurgiu, S. Jastrzebski, B. Morrone, Q. De Laroussilhe, A. Gesmundo, M. Attariyan, and S. Gelly, "Parameter-efficient transfer learning for NLP," in *Proc. Int. Conf. Mach. Learn.*, 2019, pp. 2790–2799.

[29] D. Lepikhin, H. Lee, Y. Xu, D. Chen, O. Firat, Y. Huang, M. Krikun, N. Shazeer, and Z. Chen, "GShard: Scaling giant models with conditional computation and automatic sharding," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2021, pp. 1–23.

[30] B. Zoph, I. Bello, S. Kumar, N. Du, Y. Huang, J. Dean, N. Shazeer, and W. Fedus, "ST-MoE: Designing stable and transferable sparse expert models," 2022, *arXiv:2202.08906*, doi: 10.48550/arXiv.2202.08906.

[31] Y. Zhou, T. Lei, H. Liu, N. Du, Y. Huang, V. Zhao, A. Dai, Z. Chen, Q. V. Le, and J. Laudon, "Mixture-of-experts with expert choice routing," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 35, 2022, pp. 7103–7114.

[32] J. Zhang, Y. Xie, Y. Xia, and C. Shen, "Attention residual learning for skin lesion classification," *IEEE Trans. Med. Imag.*, vol. 38, no. 9, pp. 2092–2103, 2019, doi: 10.1109/TMI.2019.2893944.

[33] J. Kawahara, S. Daneshvar, G. Argenziano, and G. Hamarneh, "Multi-resolution-tract CNN with hybrid pretrained and skin-lesion trained layers," in *Machine Learning in Medical Imaging (MLMI 2016)* (Lecture Notes in Computer Science), vol. 10019. Cham, Switzerland: Springer, 2016, pp. 164–171, doi: 10.1007/978-3-319-47157-0_20.

[34] N. C. F. Codella, D. Gutman, M. E. Celebi, B. Helba, M. A. Marchetti, S. W. Dusza, A. Kalloo, K. Liopyris, N. Mishra, H. Kittler, and A. Halpern, "Skin lesion analysis toward melanoma detection: A challenge at the 2017 international symposium on biomedical imaging (ISBI), hosted by the international skin imaging collaboration (ISIC)," in *Proc. IEEE 15th Int. Symp. Biomed. Imag. (ISBI)*, Apr. 2018, pp. 168–172.

[35] P. Tschandl, C. Rosendahl, and H. Kittler, "The HAM10000 dataset, a large collection of multi-source dermatoscopic images of common pigmented skin lesions," *Sci. Data*, vol. 5, 2018, Art. no. 180161, doi: 10.1038/sdata.2018.161.

[36] W.-L. Chiang, Z. Li, Z. Lin, Y. Sheng, Z. Wu, H. Zhang, L. Zheng, S. Zhuang, Y. Zhuang, J. E. Gonzalez, I. Stoica, and E. P. Xing. (2023). *Vicuna: An Open-Source Chatbot Impressing GPT-4 With 90%* ChatGPT Quality*. [Online]. Available: https://lmsys.org/blog/2023-03-30-vicuna/

[37] P. Koehn, "Statistical significance tests for machine translation evaluation," in *Proc. Conf. Empirical Methods Natural Lang. Process.*, 2004, pp. 388–395.

[38] L. Zhou, Y. Xu, Y. Wang, X. Chen, L. Wang, W. Chen, and C. Chen, "MABNSleepNet: A multi-view self-attention architecture based on brain network for neonatal sleep staging," *Biomed. Signal Process. Control*, vol. 118, 2026, Art. no. 109651, doi: 10.1016/j.bspc.2026.109651.
[39] X. Chen, Y. Zhang, Z. Tang, H. Wu, C. Chen, and W. Chen, "MtRBD: Advancing iRBD analysis with multi-task learning for joint sleep staging and RSWA detection," *IEEE J. Biomed. Health Informat.*, vol. 29, no. 12, pp. 8743–8750, Dec. 2025.
[40] X. Chen, X. Dong, Q. Chen, C. Chen, W. Chen, H. Chen, and B. Yin, "Application of artificial intelligence in physiological measurement based screening and diagnosis of chronic obstructive pulmonary disease: A review," *Expert Syst. Appl.*, vol. 293, Dec. 2025, Art. no. 128694.



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